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Multisubstituted benzo[*b*]furans through a copper- and/or palladium-catalyzed assembly and functionalization process

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ABSTRACT

Full details as well as the study of the scope, limitations, and further elaboration of a straightforward approach to the synthesis of 2,5,7-trisubstituted benzo[*b*]furans from 2-bromo- and 2-chloro-6-iodo-4-substituted phenols through a consecutive copper- and/or palladium-catalyzed assembly and functionalization process is described. Functionalization at the C(7) position is carried out by Suzuki–Miyaura cross-coupling, alkynylation, alkenylation, and C–N bond forming reactions. A one-pot protocol for the synthesis of 2,5,7-trisubstituted benzo[*b*]furans is also reported.

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1. Introduction

The development of methods for the construction of fused furan ring systems is of current interest.[1] This interest is stimulated by the identification of the benzofuran scaffold as a 'privileged structure' in medicinal chemistry.[2] Benzo[*b*]furans have been isolated from different medicinal plants and they were the subject of various investigations related to their therapeutic potential.[3] Members of this class of compounds have shown biological activities ranging from anti-inflammatory, anti-arrhythmic, hemostatic, antifungal, and antimicrobial, to anti-viral, anti-tumor, and anti-oxidant activities and antagonists for the H_3 receptor and angiotensin II.[4] Some of them are promising drugs against Parkinson's[5] and Alzheimer's disease.[6] Thus, these compounds are promising for discovery of new leads via pharmacophore modeling and QSAR analysis.[7] A variety of methods have been developed to prepare specific structures characterized by given pharmacological qualities. A sulfated benzofuran library of 15 sulfated monomers and 13 sulfated dimers with different charged, polar, and hydrophobic substituents was studied as the first examples of potent, small allosteric inhibitors of thrombin.[8] A novel series of 2,5,7-trisubstituted benzofuran AChE inhibitors with significant inhibitory effects on Aβ aggregation was identified. The structure–activity relationships provide a suitable template whose key structural features will contribute by optimization to the inhibitory potency of the benzofuran derivatives against AChE.[9] Usually, different types of substitution patterns in these heterocycles provide new opportunities for drug discoveries and a fine tuning of their physical properties for other applications in material science. Some benzofuran derivatives are attractive as organic materials, due to their optical and electronic properties.[10] For their potential applications, development of novel synthetic strategies is in strong demand.

In recent years, transition metal-catalyzed reactions have played a remarkable role in this area.[11–m] Particularly, palladium-[11] and copper[12]-catalyzed reactions have been widely employed in the construction of the benzo[*b*]furan core providing functional group tolerance, simplified procedures, and improved yields.

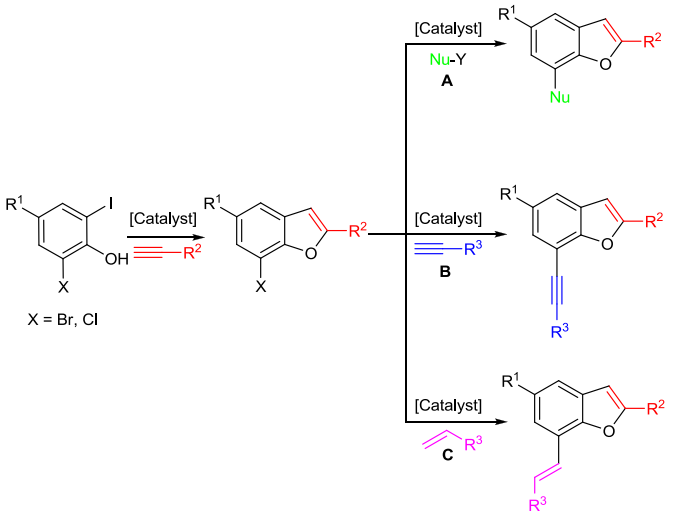
However, procedures for the synthesis of 2,5,7-trisubstituted benzo[*b*]furans remain scarcely described in the literature[9,13] and the development of simple and general methods for their preparation is a subject of great interest.

As part of our studies devoted to the construction of the benzo[*b*]furan skeleton through tandem Sonagashira coupling/5-*endo*-*dig* cyclization starting from alkynes and *o*-halophenols,[14] we observed and previously communicated[15] that readily available 2-bromo- and 2-chloro-6-iodo-4-substituted phenols can be used

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as suitable building blocks for the target 2,5,7-trisubstituted benzo[*b*]furans through a consecutive process that involves the assembly of the benzo[*b*]furan ring followed by the functionalization at the C(7) position via Suzuki–Miyaura cross-coupling and C–N bond forming reactions (Scheme 1A).

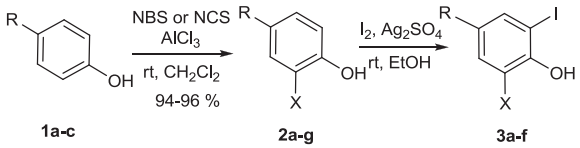


Scheme 1. Synthesis of 2,5,7-trisubstituted benzo[*b*]furans from 2-halo-6-iodo-4-substituted phenols.

Now, we wish to report full details of the results we have obtained as well as the study of the scope, limitations, and further elaboration of this approach (Scheme 1B,C) to highly functionalized benzo[*b*]furans. Moreover, a one-pot double-coupling procedure to afford a convergent synthesis of the target polysubstituted derivatives will be described.

2. Results and discussion

The starting building blocks 2-bromo- and 2-chloro-6-iodo-4-substituted phenols **3a**–**g** were prepared in high yields from commercially available 4-substituted phenols **1a**–**d** according to the sequence outlined in Scheme 2.[16]

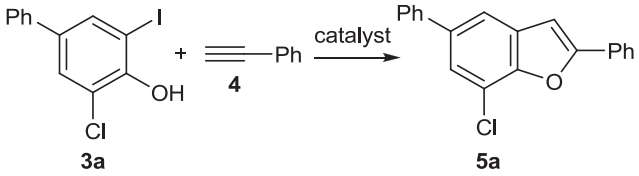


1a: R = Ph
1b: R = CH_3
1c: R = F
1d: R = $(CH_2)_2COOMe$

3a: R = Ph, X = Cl (99 %)
3b: R = Ph, X = Br (98 %)
3c: R = CH_3, X = Cl (97 %)
3d: R = CH_3, X = Br (98 %)
3e: R = F, X = Cl (99 %)
3f: R = F, X = Br (98 %)
3g: R = $(CH_2)_2CO_2Me$, X = Cl (97 %)

Scheme 2. Synthesis of 2-bromo- and 2-chloro-6-iodo-4-substituted phenols **3**.

Subsequent studies were directed toward searching for the best conditions for their conversion into the corresponding benzo[*b*]furan derivatives **5** by the reaction with terminal alkynes. 2-Chloro-6-iodo-4-phenylphenol **3a** and phenylacetylene **4** were selected as the model system (Scheme 3).



Scheme 3. Synthesis of 7-chloro-2,5-diphenylbenzo[*b*]furan **5a**.

Some results of our screening summarized in Table 1 show that benzo[*b*]furan formation was not observed by using $PdCl_2(PPh_3)_2$, CuI, and Et_2NH in DMF at room temperature (entry 1). Increasing the reaction temperature to 50 °C gave **5a** only in moderate yield (entry 2). A similar result was obtained with NH(Pr-*i*)$_2$ as the base

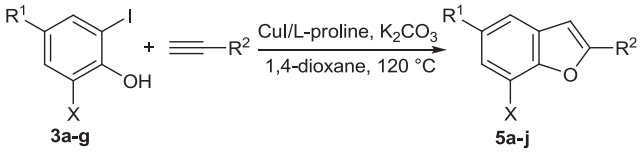
Table 1
Optimization of the reaction conditions for the synthesis of **5a** from **3a**[a]

Entry	Catalytic system (equiv)	Solvent	Base	Temperature (°C)	Time (h)	Yield % of **5a**[b]
1	$PdCl_2(PPh_3)_2$ (0.02)/ CuI (0.04)	DMF	Et_2NH	rt	24	—
2	$PdCl_2(PPh_3)_2$ (0.02)/ CuI (0.04)	DMF	Et_2NH	50	48	53
3	$PdCl_2(PPh_3)_2$ (0.02)/ CuI (0.04)	DMF	NH(Pr-*i*)$_2$	50	48	50
4	$Pd_2(dba)_3$ (0.025), $BF_4HP(Bu\text{-}t)_3$ (0.05)	DMSO	Cs_2CO_3	60	48	68
5	$Cu(phen)(PPh_3)_2]NO_3$ (0.10)	Toluene	K_2CO_3	110	2	73
6	$Cu(phen)(PPh_3)_2]NO_3$ (0.10)	Toluene	Cs_2CO_3	110	2	74
7	$Cu(phen)(PPh_3)_2]NO_3$ (0.10)	Toluene	K_3PO_4	110	2	73
8	CuI (0.10)/L-proline (0.20)	Toluene	K_3PO_4	120	1	75
9	CuI (0.10)/L-proline (0.20)	DMF	K_2CO_3	120	1	68
10	CuI (0.10)/L-proline (0.20)	DMSO	K_2CO_3	120	1	69
11	CuI (0.10)/L-proline (0.20)	1,4-Dioxane	K_2CO_3	120	2	70
12	CuI (0.10)/L-proline (0.20)	1,4-Dioxane	K_3PO_4	120	2	73
13	CuI (0.10)/L-proline (0.20)	1,4-Dioxane	Cs_2CO_3	120	1	75

[a] Reactions were carried out on a 2 mmol scale of **3a** using 6 mL of solvent, 1.5 equiv of phenylacetylene, and 2 equiv of base.
[b] Yields refer to single run and are for pure isolated products.

(entry 3). Interestingly, **5a** was isolated in good yield when $Pd_2(dba)_3$ was used as the Pd(0) source in the presence of the bulky P(Bu-*t*)$_3$ ligand generated in situ from the robust $BF_4HP(Bu\text{-}t)_3$ salt[17] under copper-free conditions (entry 4). Best results were observed by using copper catalysis. Recently, copper-catalyzed alkynylation of aryl halides has gained a growing attention due to the lower toxicity and cost of copper catalysts. A number of ligands have been examined for this type of reaction.[18] We initially used the Cu(I) complex $Cu(phen)(PPh_3)_2]NO_3$ developed by Venkataraman et al. and applied to the synthesis of 2-arylbenzo[*b*]furans.[18j,19] Under their conditions **5a** was isolated in good yields (entries 5–7). Nevertheless, we subsequently found that similar results could be obtained by using the simpler CuI/L-proline combination developed by Ma et al. and successfully employed in the N-arylation of aryl bromides[20] and in the synthesis of 2-arylindoles from terminal alkynes and 2-bromoalkynyltrifluoroacetanilides.[21] 1,4-Dioxane, toluene, and DMF were suitable solvents (entries 8–10) with this catalyst system and K_2CO_3 or K_3PO_4 or Cs_2CO_3 were effective as bases (entries 11–13).

The CuI/L-proline catalytic system was then used when we explored of 2-halo-6-iodo-4-substituted phenols **3a**–**g** with a variety of terminal alkynes (Scheme 4).



Scheme 4. Synthesis of Synthesis of 7-halo-2-phenyl-5-substituted benzo[*b*]furans **5**.

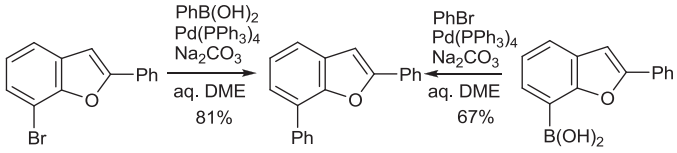
We found that 2-halo-6-iodo phenols bearing at C(4) position either an electron-releasing group or an electron-withdrawing group reacted with terminal alkynes to give the corresponding benzo[*b*]furans **5a**–**g** in good yield (Table 2). Notably, the extension of this copper-catalyzed procedure to alkyl acetylenes, also, provided the expected 2-alkyl substituted benzofurans under our conditions overcoming some previous described limitations (Table 2, entries 7 and 8).[12g]

Table 2
Synthesis of 7-halo-2-phenyl-5-substituted benzo[*b*]furans **5a**–**j**[a]

Entry	R^1	R^2	X	*t* (h)	Yield % of **5**[b]
1	Ph	Ph	Cl	2	**5a** (73)
2	Ph	Ph	Br	2	**5b** (73)
3	Me	Ph	Cl	1	**5c** (75)
4	Me	Ph	Br	1	**5d** (70)
5	F	Ph	Cl	2	**5e** (65)
6	F	Ph	Br	8	**5f** (65)
7	Ph	*n*-C_5H_{11}	Cl	3	**5g** (73)
8	Me	$(CH_2)_2$Ph	Cl	7	**5h** (60)
9	Me$(CH_2)_2CO_2$	Ph	Cl	4	**5i** (65)
10	Me	4-CH_3O–C_6H_4	Br	8	**5j** (65)
11	Ph	MeC(OH)Ph	Br	2	**5k** (85)

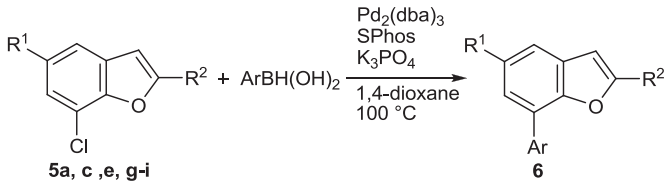
[a] Reactions were carried out on a 2 mmol scale of **3** using 6 mL of toluene at 120 °C, 1.5 equiv of alkyne, 0.10 equiv of CuI, 0.20 equiv of L-proline, and 2 equiv of K_3PO_4.
[b] Yields refer to single run and are for pure isolated products.

The potential of the functionalization at C(7) position of derivatives **5a**–**g** was next studied. Firstly, it appeared of interest to us to investigate the scope and generality of a route to 2,5,7-trisustituted benzo[*b*]furans bearing an aryl substituent at the C(7) position based on the Suzuki–Miyaura cross-coupling. To the best of our knowledge, only one example of this chemistry was reported by Moody et al.[22] who showed that 2,7-diphenylbenzo[*b*]furan could be prepared either directly by the reaction of phenyl boronic acid with 2-phenyl-7-bromobenzo[*b*]furan or indirectly by conversion of 2-phenyl-7-bromobenzo[*b*]furan into the corresponding boronic acid followed by reaction with bromobenzene (Scheme 5). The preparation of 7-benzo[*b*]furan-7-yl boronic acids involves the utilization of butyllithium, a synthetic approach that has important deficiencies in terms of flexibility and functional group tolerance when a high degree of functionalization on the benzo[*b*]furan ring is required.



Scheme 5. Synthesis of 2,7-diphenylbenzo[*b*]furan.

To overcome such drawbacks and extend the cross-coupling approach to 7-chlorobenzo[*b*]furans, our studies were directed toward the use of derivatives **5a**,**c**,**e**,**g**–**i** as coupling partners of boronic acids (Scheme 6).



Scheme 6. Palladium-catalyzed cross-coupling of 7-chlorobenzo[*b*]furan derivatives with arylboronic acids.

After a brief optimization study, we found that under the same conditions used for the preparation of 2-arylhydroxytyrosol derivatives,[16] 7-chlorobenzo[*b*]furans **5a**,**c**,**e**,**g**–**i** could give the corresponding 7-arylbenzo[*b*]furans **6** in almost quantitative yields. As shown by Table 3, the reaction can be applied to a variety of neutral, electron-rich, and electron-poor arylboronic acids. Heteroarylboronic acids can also be successfully used.

Table 3
Synthesis of 7-aryl-2-phenyl-5-substituted benzo[*b*]benzofurans **6** from 7-chlorobenzo[*b*]furans **5** and arylboronic acids[a]

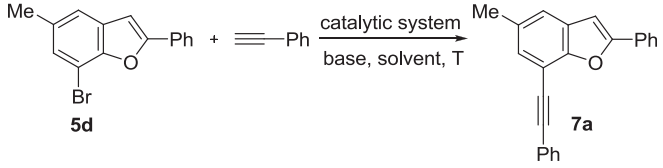
Entry	R^1	R^2	Ar	*t* (h)	Yield % of **6**[b]
1	Ph	Ph	3-$CH_3O_2CC_6H_4$	2	**6aa** (99)
2	Ph	Ph	4-$CH_3OC_6H_4$	3	**6ab** (99)
3	Ph	Ph	4-$C_6H_5C_6H_4$	2	**6ac** (99)
4	Ph	Ph	3-CH_3,4-FC_6H_3	3	**6ad** (99)
5	Ph	Ph	3-Thienyl	4	**6ae** (99)
6	Me	Ph	3-$CH_3O_2CC_6H_4$	2	**6ca** (95)
7	Me	Ph	4-$CH_3OC_6H_4$	3	**6cb** (95)
8	Me	Ph	4-$C_6H_5C_6H_4$	2	**6cc** (99)
9	Me	Ph	3-CH_3,4-FC_6H_3	3	**6cd** (95)
10	Me	Ph	3-Thienyl	3	**6ce** (99)
11	F	Ph	3-$CH_3O_2CC_6H_4$	2	**6ea**(99)
12	F	Ph	4-$CH_3OC_6H_4$	3	**6eb** (95)
13	F	Ph	4-$C_6H_5C_6H_4$	2	**6ec** (99)
14	F	Ph	3-CH_3,4-FC_6H_3	2	**6ed** (95)
15	F	Ph	3-Thienyl	3	**6ee** (98)
16	Ph	*n*-C_5H_{11}	4-$CH_3COOC_6H_4$	4	**6ga** (99)
17	Ph	*n*-C_5H_{11}	3-CH_3,4-FC_6H_3	4	**6gb** (99)
18	Ph	*n*-C_5H_{11}	4-$C_6H_5C_6H_4$	2	**6gc** (99)
19	Ph	*n*-C_5H_{11}	3-Thienyl	5	**6gd** (99)
20	Ph	*n*-C_5H_{11}	C_6H_5	3	**6ge** (99)
21	Me	$(CH_2)_2$Ph	4-$CH_3COOC_6H_4$	1.5	**6ha** (94)
22	Me	$(CH_2)_2$Ph	3-CH_3,4-FC_6H_3	3	**6hb** (99)
23	Me	$(CH_2)_2$Ph	4-$C_6H_5C_6H_4$	2	**6hc** (99)
24	Me	$(CH_2)_2$Ph	3-Thienyl	3	**6hd** (99)
25	Me	$(CH_2)_2$Ph	C_6H_5	1.5	**6he** (97)
26	Me$(CH_2)_2CO_2$	Ph	3-$CH_3COOC_6H_4$	3	**6ia** (99)
27	Me$(CH_2)_2CO_2$	Ph	4-$CH_3OC_6H_4$	3	**6ib** (99)[c]
28	Me$(CH_2)_2CO_2$	Ph	4-$C_6H_5C_6H_4$	24	**6ic** (99)
29	Me$(CH_2)_2CO_2$	Ph	3-CH_3,4-FC_6H_3	8	**6id** (99)
30	Me$(CH_2)_2CO_2$	Ph	3-Thienyl	12	**6ie** (99)

[a] Reactions were carried out on a 0.4 mmol scale of **5** using 2 mL of 1,4-dioxane at 100 °C, 1.5 equiv of arylboronic acids, 0.04 equiv of $Pd_2(dba)_3$, 0.04 equiv of SPhos, and 3 equiv of K_3PO_4.
[b] Yields refer to single run and are for isolated products.
[c] Reaction temperature=120 °C.

Secondly, the palladium-catalyzed functionalization of 7-bromobenzo[*b*]furans was applied to the synthesis of alkynyl/alkenyl C(7) functionalized benzo[*b*]furans through Sonogashira/Heck reactions. Sonogashira-type cross-coupling of 2,3-dibromobenzofuran and 2,3,5-tribromobenzofuran was reported to smoothly proceed only at C(2) position.[23]

One current challenge associated with the Sonogashira coupling reaction is its inefficiency when 'unreactive' aryl halides are employed.[24] As illustrated in Table 4, we failed to obtained the alkynyl derivative **7** under copper catalysis (entry 1) and unsatisfactory results from synthetic point of view were observed

Table 4
Synthesis of 5-methyl-2-phenyl-7-(phenylethynyl)benzo[*b*]furan **7**[a]



Entry	Catalytic system	Base	Solvent/*T* (°C)	*t* (h)	Yield % of **7**[b]
1	CuI/L-proline	Cs_2CO_3	1,4-Dioxane/110	24	—
2	$PdCl_2(PPh_3)_2$/CuI	*i*-Pr_2–NH	1,4-Dioxane/100	24	35
3	$Pd_2(dba)_3$/P(*c*-$C_6H_{11})_3$	Cs_2CO_3	1,4-Dioxane/100	24	20
4	$Pd_2(dba)_3$/[(*t*-Bu)$_3$HP]$_3BF_4$	Cs_2CO_3	1,4-Dioxane/60	6	99

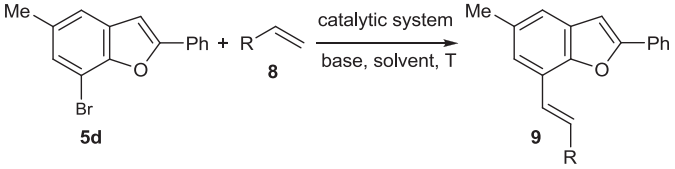
[a] Reactions were carried out on a 0.34 mmol of **5c** scale using 1 mL of 1,4-dioxane at 60 °C.
[b] Yields refer to single run and are for isolated products.

both in standard Sonogashira conditions (Table 4, entry 2) and under copper-free conditions by using $Pd_2(dba)_3$/P(*c*-$C_6H_{11})_3$ as catalyst.[25] The steric bulk and electron-rich P*t*-Bu_3 had a pronounced impact on the process accomplishing the formation of the expected alkynyl derivative in very high yield (entry 4).[26] The most effective base among those surveyed was Cs_2CO_3.

Having optimized the reaction conditions for the Sonogashira reaction of **5d**, we determined the substrate scope of the process. As shown in Table 5, in general the desired products **7** were obtained in good to excellent yields with both aryl and alkyl acetylenes. Relative lower yields occurred in the cross-coupling reactions of **5j**,**f**,**d** with the tertiary propargylic alcohol 2-methylbut-3-yn-2-ol (Table 5, entries 5–7).

Although 2,3-dialkenylbenzofurans and functionalized dibenzofurans were prepared based on domino twofold Heck/6π-electrocyclization reactions of 2,3-di-and 2,3,5-tribromobenzofuran,[27] to the best of our knowledge the alkenylation reaction at the C(7) position has not been previously reported.

The Heck coupling of the deactivated bromide derivative **5d** with various alkenes **8** was then performed (Scheme 7).



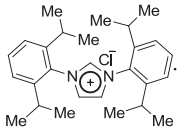
Scheme 7. Palladium-catalyzed cross-coupling of 7-bromobenzo[*b*]furan **5c** with alkenes.

The Heck coupled derivative **9a** was obtained in 63% yield by reacting **5c** with styrene in 1,4-dioxane at 110 C° in the presence of $Pd_2(dba)_3$/[(*t*-Bu)$_3$HP]$_3BF_4$ as the catalytic system and Cs_2O_3 as the base. The use of the Hermann catalyst and MeCOOK as a base gave inferior results under the same conditions (Table 6, entry 1). It has been recognized that palladium complexes derived from sterically bulky electron-rich phosphines,[28] *N*-heterocyclic carbenes,[29] and carbocyclic carbenes[30] facilitate the Heck reaction of deactivated aryl/heteroaryl halides. Surprisingly, we failed to obtain efficiently the desired cross-coupling derivatives with ethyl or butyl acrylate (Table 6, entry 3–8). An unusual propensity of substrate **5d** to protodebrominate in the presence of acrylates caused the failure of the Heck coupling.

Table 6
Optimization of the reaction conditions for the Heck reaction of 5-methyl-2-phenyl-7-bromobenzofuran **5d**[a]

Entry	R	Catalytic system (equiv)	Yield % of **9**[b]
1[c]	Ph	Hermann catalyst	**9a** (34)
2[d,e]	Ph	$Pd_2(dba)_3$/[(*t*-Bu)$_3$HP]$_3BF_4$	**9a** (63)
3[c,f]	MeCOO	Hermann catalyst	**9b** (–)
4[c,f]	BuCOO	PdEnCat	**9c** (–)
5[f,g]	BuCOO	$Pd(OAc)_2$/L[h]	**9c** (–)
6[c,f]	BuCOO	$Pd(OAc)_2$/L[h]	**9c** (–)
7[c,f]	MeCOO	Hermann catalyst	**9b** (12)
8[d,e]	MeCOO	$Pd_2(dba)_3$/[(*t*-Bu)$_3$HP]$_3BF_4$	**9b** (–)

[a] Reactions were carried out in 24 h on a 0.34 mmol scale of **5** in 1 mL of DMF at 110 °C, using 1.5 equiv of alkene, 1.1 equiv of base, 0.02 equiv of catalyst.
[b] Yields refer to single run and are for isolated products.
[c] Using AcOK as base.
[d] In 1,4-dioxane.
[e] Using Cs_2CO_3 as base.
[f] At 120 °C.
[g] Using K_2CO_3/Cs_2CO_3 as combination of bases.
[h] L=[e] L =



Results of further investigations demonstrated that all substrates tested were tolerated in this Heck coupling with good yields (Table 7).

The potential of our strategy for the preparation of 2,5,7-trisubstituted benzo[*b*]furans is further demonstrated by the formation of 7-amino derivatives **10**. The results of a brief screening aimed to the optimization of the reaction conditions for C–N bond

Table 5
Synthesis of 2,5-disubstituted-7-alkynylbenzo[*b*]furan **7**[a]

Entry	**5**	Alkyne	*t* (h)	Yield % of **7**[b]
1	**5d**	—Ph	6	**7a** (99)
2	**5f**	—C_6H_4-OMe	3	**7b** (82)
3	**5d**	—C_6H_4-OMe	3	**7c** (83)
4	**5j**	—Ph	2	**7d** (80)
5	**5j**	—C(OH)Me_2	2	**7e** (50)
6	**5f**	—C(OH)Me_2	6	**7f** (50)
7	**5d**	—C(OH)Me_2	6	**7g** (60)
8	**5k**	—Ph	1	**7h** (99)
9	**5k**	—C_5H_{11}	24	**7i** (79)

[a] Reactions were carried out on a 0.34 mmol scale of **5** using 1 mL of 1,4-dioxane at 60 °C, using 1.5 equiv of alkyne, 0.02 equiv of $Pd_2(dba)_3$, 0.08 equiv of [(*t*-Bu)$_3$HP]$_3BF_4$, 1.1 equiv of Cs_2CO_3.
[b] Yields refer to single run and are for isolated products.

Table 7
Synthesis of 2,5-disubstituted-7-styrylbenzo[*b*]furan **9**[a]

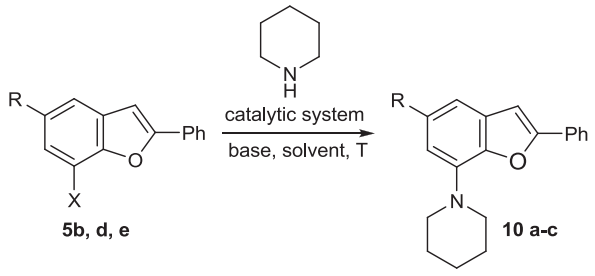
Entry	**5**	*t* (h)	Yield % of **9**[b]
1	**5d**	49	**9a** (63)
2	**5f**	24	**9b** (70)
3	**5j**	48	**9c** (72)

[a] Reactions were carried out in 24 h on a 0.34 mmol scale of **5** in 1 mL of 1,4-dioxane at 110 °C, using 1.5 equiv of styrene, 1.1 equiv of Cs_2CO_3, 0.02 equiv of $Pd_2(dba)_3$, 0.08 equiv of $[(t\text{-}Bu)_3HP]_3BF_4$.
[b] Yields refer to single run and are for isolated products.

formation by reacting 7-bromo[*b*]furans **5b**, **d**, **e** with piperidine are summarized in Table 8.

The best results were obtained using the $Pd_2(dba)_3$/X-Phos catalytic system.[31] Copper based protocols[32] resulted less efficient in promoting the C–N bond formation at the C-7 position.

Table 8
Optimization of the reaction conditions for the synthesis of 7-amino-2,5-disubstituted-benzo[*b*]furans **10**[a]



5b, d, e **10 a-c**

Entry	**5**	Catalytic system (equiv)	Time (h)	Yield % of **10**[b]
1[c]	**5b**	CuI/L-proline	72	**10a** (55)
2[d,e,f,g]	**5b**	$Pd_2(dba)_3$/X-Phos	24	**10a** (71)
3[c,h]	**5d**	CuI/L-proline	72	**10b** (46)
4[c]	**5d**	CuI/L-proline	36	**10b** (56)
5[h]	**5d**	CuI/L-proline	72	**10b** (27)
6[c]	**5d**	CuI/*N*-methylglycine	36	**10b** (55)
7[i]	**5d**	CuI/*N*,*N*-dimethylglycine	72	**10b** (39)
8[d,e,f,g]	**5d**	$Pd_2(dba)_3$/X-Phos	12	**10b** (99)
9[d,e,f,g]	**5e**	$Pd_2(dba)_3$/X-Phos	12	**10c** (72)

[a] Reactions were carried out in 24 h on a 0.34 mmol scale using 1 mL of solvent at 120 °C, using 10 equiv of piperidine, 2 equiv of K_2CO_3, 0.10 equiv of CuI, and 0.20 equiv of N-ligand catalyst.
[b] Yields refer to single run and are for isolated products.
[c] In DMSO.
[d] In toluene.
[e] Using NaOBu-*t*.
[f] At 80 °C.
[g] With 0.025 equiv of $Pd_2(dba)_3$ and 0.10 equiv of X-Phos.
[h] At 100 °C in 1,4-dioxane.
[i] At 120 °C in 1,4-dioxane.

Using the 'optimal' condition found, we next examined the reaction using 7-chlorobenzo[*b*]furan derivatives in order to determine the scope and limitations of the palladium-catalyzed C–N bond formation process. The results are listed in Table 9. To our delight, the cross-coupling reaction of the less reactive chlorides with secondary amines proceeds as well as with the corresponding 7-bromoderivatives.

Moreover, the reaction of the 7-chloro-2,5-disubstituted benzo[*b*]furans with *N*,*N*-dialkylhydrazines occurs in high to excellent yields.

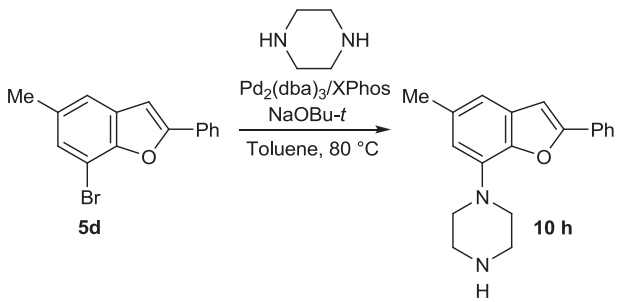
Interestingly, unprotected piperazine can be used in the coupling reaction with **5d** to give **10h** in 60% yield although mono-Boc derivatives have been suggested to be required to circumvent bis-arylation reactions.[33] Compound **10h** was reported to form only in 20% yield upon treatment of 7-bromobenzo[*b*]furan with *N*-Boc-piperazine in the presence of $Pd(dba)_2$, P(*o*-tol)$_3$, and NaOBu-*t* in toluene at 100 °C followed by a deprotection step with trifluoroacetic acid (Scheme 8).

Table 9
Synthesis of 7-aminobenzo[*b*]benzofurans **10** from 7-chlorobenzo[*b*]furans **5** and secondary amines/hydrazines[a]

Entry	**5**	R_2NH or $R_2N–NH_2$	Time (h)	Yield % of **10**[b]
1	**5b**		12	**10a** (71)
2	**5c**		3	**10b** (60)
3	**5c**		1.3	**10c** (86)
4	**5c**		1	**10d** (76)
5	**5c**		2.5	**10e** (56)
6	**5c**	H_2NNMe_2	3.5	**10f** (61)
7	**5c**		1.5	**10g** (99)
8	**5d**		24	**10b** (*99*)
9	**5d**		40	**10h** (*60*)
10	**5e**		12	**10i** (86)
11	**5g**		5	**10j** (55)
12	**5f**		12	**10i** (71)
13	**5g**		3	**10k** (70)
14	**5g**	H_2NNMe_2	2	**10l** (90)
15	**5g**		2	**10m** (91)
16	**5g**		3	**10n** (65)
17	**5j**		1	**10o** (83)
18	**5k**	H_2NNMe_2	7	**10p** (67)

[a] Reactions were carried out on a 0.34 mmol scale of **5** in 1 mL of toluene at 80 °C, 10 equiv of nitrogen compound, 0.025 equiv of $Pd_2(dba)_3$, 0.10 equiv of X-Phos, and 2 equiv of NaOBu-*t*.
[b] Yields refer to single run and are for isolated products.

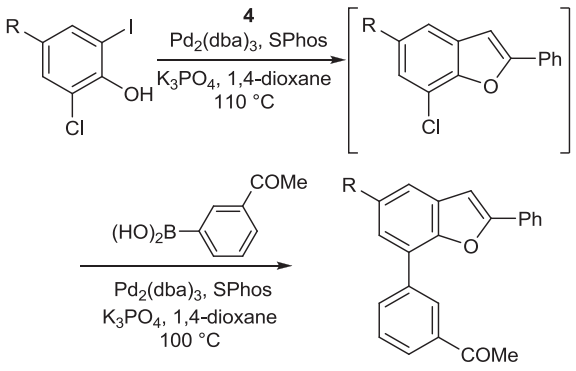
Having successfully established a protocol to produce 2,5,7-trisubstituted benzo[*b*]furans through the consecutive process



5d **10 h**

Scheme 8. Palladium-catalyzed amination of 7-bromobenzo[*b*]furans **5d** with piperazine.

involving the assembly of the benzo[*b*]furan ring followed by the functionalization at the C(7) position via Suzuki–Miyaura cross-coupling and C–N bond forming reactions, we examined the versatility of a one-pot preparation. One-pot multistep processes are economically advantageous considering the outlays for the catalyst, solvent, purification materials, and time. We used the same catalytic system for the two successive reaction steps in which at first the terminal alkyne would react selectively with 2-chloro-6-iodo-4-substituted phenols **3a**,**c**,**e** at the more reactive C-6 position to give the corresponding 7-chloro-2,5-disubstituted benzo[*b*]furan derivative **5**. Then the boronic acid would be added to react at the C-7 position (Scheme 9).



Scheme 9. Palladium-catalyzed one-pot assembly of 2,5,7-trisubstituted benzo[*b*]furans.

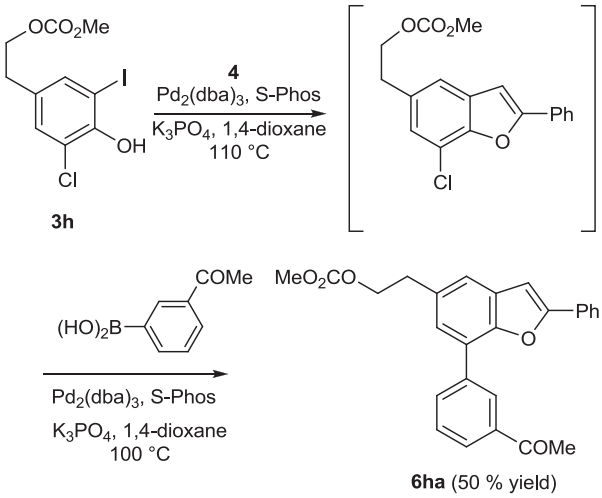
The best reaction conditions were the following: the starting 2-chloro-6-iodo-4-substituted phenol was treated with phenylacetylene in the presence of $Pd_2(dba)_3$, SPhos as the catalytic system, and K_3PO_4 as the base in 1,4-dioxane at 110 °C. After the end of the reaction, an extraction was made and the boronic acid derivative (1.5 equiv) was added together with $Pd_2(dba)_3$, SPhos/K_3PO_4 to the residue and the reaction mixture was then heated at 100 °C. These sequences afforded the target 2,5,7- tri-substituted benzo[*b*]furan in satisfactory overall yield (Table 10).

Table 10
One-pot palladium-catalyzed synthesis of 2,5,7-trisubstituted benzo[*b*]furans **6**[a,b]

Entry	**3**	Time (step 1, h)	Time (step 2, h)	Yield % of **6**[c]
1	**3a**	3	4	**6aa** (50)
2	**3c**	3	4	**6ca** (60)
3	**3e**	3	6	**6ea** (50)

[a] Reactions were carried out on a 0.37 mmol scale of **3**. Reaction condition for step 1: 1.5 equiv of phenylacetylene, 0.025 equiv of $Pd_2(dba)_3$ and 0.10 equiv of SPhos, 2 equiv of K_3PO_4 in 2 mL of 1,4-dioxane at 110 °C. Reaction condition for step 2: 1.5 equiv of boronic acid, 0.02 equiv of $Pd_2(dba)_3$ and 0.04 equiv of SPhos, 1.1 equiv of K_3PO_4 in 2 mL of 1,4-dioxane at 100 °C.
[b] Between step 1 and step 2 an extraction was performed.
[c] Yields refer to single run and are for isolated products.

As part of a program devoted to the chemical valorization of widespread diffused molecules in renewable sources,[16] we extended the one-pot process to the preparation of the polysubstituted benzo[*b*]furan derivative **6ha** by starting from the 4-(2-hydroxyethyl)phenol (tyrosol), a naturally occurring bioactive molecule found in extra virgin olive oil and agricultural wastewaters.[34] The described one-pot double cross-coupling procedure was applied on substrate **3h** derived by halogenation reaction of the selectively protected tyrosol. The expected product **6ha** was isolated in 50% overall yield (Scheme 10).



Scheme 10. One-pot palladium-catalyzed synthesis the tyrosol derivative **6ha**.

One-pot two-step cross-coupling of compound **5d**, with phenylacetylene and piperidine to give the 7-amino derivative **10b** was also studied. For the optimization of this process, we screened a variety of reaction conditions to compare copper versus palladium catalysis for the one-pot protocol (Table 11). Under the best conditions developed the product **10b** was isolated in 33% overall yield by using the same copper catalyst and 50% overall yield whith the same palladium catalyst in both the reaction steps. Analogously **10b** could be conveniently prepared through a process in which the first step was catalyzed by CuI/L-proline. Subsequently, the palladium-catalyzed C-N bond formation step was carried out on the resulting crude mixture after extraction and evaporation of the solvent. Using this procedure, **10b** was isolated in the same overall yield (50%) than that obtained via the one-pot palladium-catalyzed double cross-coupling protocol.

3. Conclusion

In summary, this paper describes a straightforward general protocol for the synthesis of 2,5,7-trisubstituted benzo[*b*]furans from readily available 2-bromo- and/or 2-chloro-6-iodo-4-substituted phenols.

A comparison of the copper and palladium catalysis shows that both transition metals are suitable catalysts for the site selective sequential alkynylation/cyclization process.

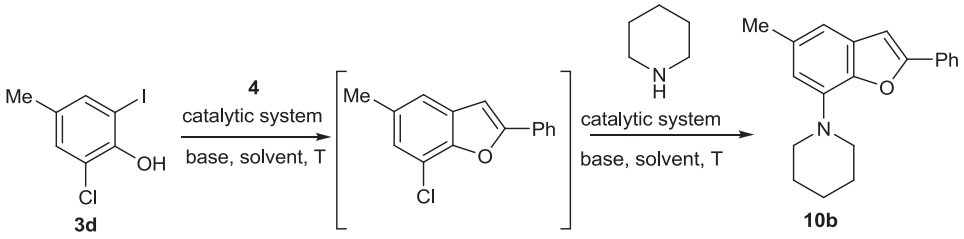
$Pd_2(dba)_3$/SPhos and $Pd_2(dba)_3$/X-Phos catalysts achieve very efficiently the consecutive functionalization at the C(7) position of both bromo- and chloro-benzo[*b*]furans through cross-coupling with boronic acids and secondary amines/*N*,*N*-dialkylhydrazines, respectively.

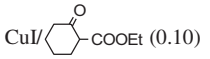
The use of $Pd_2(dba)_3$/[(*t*-Bu)$_3$HP]$_3$BF$_4$ as the catalyst and Cs_2O_3 as the base achieves the alkynylation of 7-bromo-benzo[*b*]furan derivative under copper-free conditions in 1,4-dioxane. Under these latter conditions the Heck coupling of the 7-heroaryl bromides could be, also, performed.

One-pot double cross-coupling protocols for the synthesis of polysubstituted benzo[*b*]furans from 2-bromo- and/or 2-chloro-6-iodo-4-substituted phenols can accomplish the formation of the target products in about the same overall yield than that obtained through the stepwise procedure.

Extension of the one-pot protocol can allow the preparation of one polysubstituted benzo[*b*]furans from tyrosol.

Table 11
Optimization of the reaction conditions for one-pot synthesis of **10b**[a]



Entry	Catalytic system 1 (equiv)	Catalytic system 2 [molar ratio]	Solvent	Base	T_1/T_2 (°C)	t_1/t_2 (h)	Yield % of **10d**[b]
1	CuI/L-proline (0.10:0.20)		DMSO	K_2CO_3	100:100	4:24	14
2	CuI/L-proline (0.10:0.20)		DMSO	K_2CO_3	120:100	2:24	18
3	CuI/L-proline (0.10:0.20)		1,4-Dioxane	K_2CO_3	120:100	6:24	33
4	Cu[(phen)(PPh_3)$_2$]NO_3 (0.10)		1,4-Dioxane	K_3PO_4	120:110	6:24	—
5			1,4-Dioxane	K_2CO_3	120:120	12:24	15
6			1,4-Dioxane	K_2CO_3	120:120	12:24	15
7	CuI/L-proline (0.10:0.20)	$Pd_2(dba)_3$/X-Phos (0.025:0.10)	Toluene	NaOBu-*t*	120:80	2:24	27
8	CuI/L-proline (0.10:0.20)	$Pd_2(dba)_3$/X-Phos(0.025:0.10)	Toluene	NaOBu-*t*	120:80	2:24	50[c]
9	$Pd_2(dba)_3$/X-Phos (0.025:0.10)	$Pd_2(dba)_3$/X-Phos (0.025:0.10)	Toluene	NaOBu-*t*	100:80	3:24	50[c]

[a] Reactions were carried out on a 0.37 mmol scale of **3**. Reaction condition for step 1: 1.5 equiv of phenylacetylene, 2 equiv of base in 2 mL of solvent. Reaction condition for step 2: 10 equiv of piperidine.
[b] Yields refer to single run and are for isolated products.
[c] Between step 1 and step 2 an extraction was made before the addition of palladium catalyst and base.

4. Experimental section

4.1. General conditions

All the starting materials, catalysts, bases, solvents, and eluents are commercially available and were used as purchased, without further purification. ^{1}H NMR (400.13 MHz) and ^{13}C NMR (100.6 MHz) were recorded with a Bruker Avance 400 spectrometer. Infrared spectra were recorded on a Jasco FT/IR-430 spectrophotometer. Melting points were determined with a Büchi B-545 apparatus and are uncorrected. GC–MS spectra were recorded with Shimadzu (QP 2010S), LC–MS were recorded with Thermo Finningan Analyzer Ion Trap LXQ with ESI source.

4.2. General procedures and methods

*4.2.1. General procedure for the copper-catalyzed synthesis of 7-halobenzo[b]furans (**5**): synthesis of 7-bromobromo-5-methyl-2-phenylbenzo[b]furan (**5d**).* A solution of 2-bromo-4-methyl-6-iodophenol **3d** (0.600 g, 1.92 mmol), CuI (0.036 g, 0.192 mmol), L-proline (0.044 g, 0.384 mmol), and K_3PO_4 (0.815 g, 3.84 mmol) in toluene (6 mL) was added phenylacetylene **4** (0.309 mL, 2.88 mmol) and the mixture was stirred at 120 °C for 1 h. After cooling, the reaction mixture was dried under reduced pressure and the residue was purified by flash chromatography (SiO_2, 50 g; *n*-hexane/ethyl acetate 98:2 v/v) to give **5d** (0.387 g, 70%) as a white solid. Mp: 70–71 °C. MS (relative intensity): *m*/*z* 288 (M^+, 100), 286 (M^+, 100), 207 (52), 178 (46); IR (KBr): 3112, 3033, 2927, 1459, 761 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.91 (d, *J*=7.6 Hz, 2H), 7.48–7.41 (m, 2H), 7.39 (t, *J*=7.2 Hz, 1H), 7.38 (s, 1H), 7.31 (s, 1H), 7.02 (s, 1H), 2.45 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 156.7, 150.4, 134.1, 130.5, 130.0, 128.9, 128.8, 128.3, 125.1, 120.0, 130.2, 101.6, 21.1. Anal. Calcd for $C_{15}H_{11}BrO$: C, 62.74; H, 3.86; Br, 27.83. Found: C, 62.66; H, 3.87; Br, 27.88.

*4.2.2. Procedure for the palladium-catalyzed cross-coupling of 7-chlorobenzo[b]furans (**5a**,**c**,**e**) with arylboronic acids: synthesis of 7-(3-carbomethoxyphenyl)-5-methyl-2-phenylbenzo[b]furan (**6ca**).* A solution of 2-phenyl-5-methyl-7-chlorobenzo[*b*]furan **5c** (0.100 g, 0.41 mmol), 3-carbomethoxyphenyl boronic acid (0.112 g, 0.62 mmol), $Pd_2(dba)_3$ (0.0075 g, 0.0082 mmol), SPhos (0.0067 g, 0.0164 mmol), and K_3PO_4 (0.261 g, 1.23 mmol) in 1,4-dioxane (2 mL) was stirred at 100 °C for 2 h. After cooling, the reaction mixture was dried under reduced pressure and the residue was purified by flash chromatography (SiO_2, 50 g; *n*-hexane/ethyl acetate 90:10 v/v) to give **6ca** (0.133 g, 95%) as a white solid. Mp: 130–131 °C. MS: *m*/*z* 365 (MNa^+); IR (KBr): 3100, 1708, 761, 690 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 8.69 (s, 1H), 8.21 (d, *J*=8.0 Hz, 1H), 8.11 (d, *J*=7.6 Hz, 1H), 7.89 (d, *J*=7.6 Hz, 2H), 7.63 (t, *J*=7.6 Hz, 1H), 7.47–7.29 (m, 5H), 7.04 (s, 1H), 4.02 (s, 3H), 2.54 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 194.3, 167.1, 156.2, 150.4, 136.9, 133.0, 132.8, 130.6, 130.4, 130.3, 129.6, 128.8, 128.7, 128.61, 128.57, 124.9, 123.5, 120.6, 101.2, 52.2, 21.4. Anal. Calcd for $C_{23}H_{18}O_3$: C, 80.68; H, 5.30. Found: C, 80.60; H, 5.31.

*4.2.3. Procedure for the palladium-catalyzed amination reaction of 7-halobenzo[b]furans (**5b**,**c**,**d**,**e**,**f**,**g**,**j**): synthesis of 1-(5-methyl-2-phenylbenzo[b]furan-7-yl)piperidine (**10b**).* A solution of 2-phenyl-5-methyl-7-bromobenzo[*b*]furan **5d** (0.100 g, 0.34 mmol), $Pd_2(dba)_3$ (0.0078 g, 0.0085 mmol), X-Phos (0.0162 g, 0.034 mmol), NaOBu-*t* (0.065 g, 0.68 mmol), and piperidine (0.337 mL, 3.4 mmol) in toluene (1 mL) was stirred at 80 °C for 24 h. After cooling, the reaction mixture was dried under reduced pressure and the residue was purified by flash chromatography (SiO_2, 50 g; *n*-hexane/ethyl acetate 85:15 v/v) to give **10b** (0.133 g, 99%) as a yellow solid. Mp 139–140 °C; MS (relative intensity): *m*/*z* 291 (M^+, 100), 235 (13), 207 (17); IR (KBr) 3100, 2921, 1596, 765, 690 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.87 (d, *J*=8.4 Hz, 2H), 7.46 (t, *J*=7.6 Hz, 2H), 7.36 (d, *J*=7.2 Hz, 1H), 6.98 (s, 1H), 6.95 (s, 1H), 6.62 (s, 1H), 3.37 (br t, *J*=5.2 Hz, 4H), 2.43 (s, 3H), 1.91–1.85 (m, 4H), 1.72–1.65 (m, 2H); C NMR ($CDCl_3$): δ 155.0, 145.2, 137.8, 133.2, 130.8, 130.3, 128.7, 128.2, 124.8, 113.3, 113.1, 101.5, 51.4, 26.2, 24.6, 21.8. Anal. Calcd for $C_{20}H_{21}NO_3$: C, 82.44; H, 7.26; N, 4.81. Found: C, 82.36; H, 7.27; N, 4.80.

*4.2.4. Procedure for the palladium-catalyzed cross-coupling of 7-bromobenzo[b]furans (**5d**,**f**,**j**) with alkynes: synthesis of 5-methyl-2-*

*phenyl-7-(phenylethynyl)benzo[b]furan (**7a**).* A solution of 2-phenyl-5-methyl-7-bromo[*b*]furan **5d** (0.100 g, 0.34 mmol), phenylacetylene **4** (0.058 mL, 0.53 mmol), $Pd_2(dba)_3$ (0.0064 g, 0.007 mmol), $[(t\text{-}Bu)_3PH]BF_4$ (0.0081 g, 0.028 mmol), and Cs_2CO_3 (0.125 g, 0.39 mmol) in 1,4-dioxane (1 mL) was stirred at 60 °C for 6 h. After cooling, the reaction mixture was dried under reduced pressure and the residue was purified by flash chromatography (SiO_2, 50 g; *n*-hexane/ethyl acetate 95:5 v/v) to give **7a** (0.107 g, 99%) as a white solid. Mp: 133–134 °C; MS: *m*/*z* 331 (MNa^+); IR (KBr): 2924, 2362, 1599, 1568, 1489, 1466, 1442 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.96 (d, *J*=7.2 Hz, 2H), 7.76–7.74 (m, 2H), 7.53–7.28 (m, 8H), 6.99 (s, 1H), 2.49 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 156.5, 153.2, 132.5, 131.9, 130.3, 129.6, 128.9, 128.7, 128.4, 125.1, 123.5, 121.5, 106.9, 101.3, 93.6, 84.3, 21.2. Anal. Calcd for $C_{23}H_{16}O$: C, 89.58; H, 5.23. Found: C, 89.65; H, 5.22.

*4.2.5. Procedure for the palladium-catalyzed Heck reaction of 7-bromobenzo[b]furans **5d** with alkenes: synthesis of 5-methyl-2-phenyl-7-(2-phenylethenyl)benzo[b]furan (**9a**).* A solution of 2-phenyl-5-methyl-7-bromo[*b*]furan **5d** (0.100 g, 0.34 mmol), styrene (0.060 mL, 0.53 mmol), $Pd_2(dba)_3$ (0.0064 g, 0.007 mmol), $[(t\text{-}Bu)_3PH]BF_4$ (0.0081 g, 0.028 mmol), and Cs_2CO_3 (0.125 g, 0.39 mmol) in 1,4-dioxane (1 mL) was stirred at 110 °C for 48 h. After cooling, the reaction mixture was dried under reduced pressure and the residue was purified by flash chromatography (SiO_2, 50 g; *n*-hexane/ethyl acetate 90:10 v/v) to give **9a** (0.068 g, 63%) as white solid. Mp: 118–120 °C. MS: *m*/*z* 333 (MNa^+); IR (KBr): 2922, 1597, 1564, 1493, 1444 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.95 (d, *J*=7.6 Hz, 2H), 7.69 (d, *J*=7.6 Hz, 3H), 7.53–7.27 (m, 9H), 7.01 (s, 1H), 2.51 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 156.0, 150.9, 137.9, 132.7, 131.6, 130.6, 129.9, 128.8, 128.7, 128.5, 127.7, 126.7, 124.9, 124.6, 123.7, 121.5, 120.1, 101.3, 21.4. Anal. Calcd for $C_{23}H_{18}O$: C, 89.00; H, 5.85. Found: C, 88.92; H, 5.86.

*4.2.6. One-pot procedure for palladium-catalyzed alkynylation/cyclization reaction and functionalization at C(7) position via Suzuki–Miyaura cross-coupling: synthesis of 7-(3-carbomethoxyphenyl)-5-methyl-2-phenylbenzo[b]furan (**6ca**).* A solution of 2-bromo-4-methyl-6-iodophenol **3d** (0.100 g, 0.37 mmol), $Pd_2(dba)_3$ (0.0085 g, 0.0093 mmol), SPhos (0.015 g, 0.037 mmol), phenylacetylene (0.047 mL, 0.44 mmol), and K_3PO_4 (0.157 g, 0.74 mmol) in 1,4-dioxane (2 mL) was stirred at 110 °C for 3 h. After cooling, the reaction mixture was diluted with Et_2O and washed with water and with brine. The organic layer was dried over Na_2SO_4, filtered, and concentrated under reduced pressure. The residue was dissolved in 1,4-dioxane (2 mL) and 3-carbomethoxyphenyl boronic acid (0.099 g, 0.55 mmol), $Pd_2(dba)_3$ (0.007 g, 0.0074 mmol), SPhos (0.0061 g, 0.0148 mmol), and K_3PO_4 (0.235 g, 1.10 mmol) were added and the obtained solution was stirred at 100 °C for 4 h. After cooling, the reaction mixture was dried under reduced pressure and the residue was purified by flash chromatography (SiO_2, 50 g; *n*-hexane/ethyl acetate 90:10 v/v) to give **6ca** (0.076 g, 60%) as a white solid. Mp: 130–131 °C. MS: *m*/*z* 365 (MNa^+); IR (KBr): 3100, 1708, 761, 690 cm^{-1}; ^1H NMR ($CDCl_3$): δ 8.69 (s, 1H), 8.21 (d, *J*=8.0 Hz, 1H), 8.11 (d, *J*=7.6 Hz, 1H), 7.89 (d, *J*=7.6 Hz, 2H), 7.63 (t, *J*=7.6 Hz, 1H), 7.47–7.29 (m, 5H), 7.04 (s, 1H), 4.02 (s, 3H), 2.54 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 194.3, 167.1, 156.2, 150.4, 136.9, 133.0, 132.8, 130.6, 130.4, 130.3, 129.6, 128.8, 128.7, 128.61, 128.57, 124.9, 123.5, 120.6, 101.2, 52.2, 21.4. Anal. Calcd for $C_{23}H_{18}O_3$: C, 80.68; H, 5.30. Found: C, 80.60; H, 5.31.

*4.2.7. One-pot for palladium-catalyzed alkynylation/cyclization reaction and functionalization at C(7) position via C–N bond forming reactions: synthesis of 1-(5-methyl-2-phenylbenzo[b]furan-7-yl)piperidine (**10b**).* A solution of 2-bromo-4-methyl-6-iodophenol **3d** (0.100 g, 0.31 mmol), $Pd_2(dba)_3$ (0.007 g, 0.0078 mmol), X-Phos (0.0148 g, 0.031 mmol), phenylacetylene (0.04 mL, 0.37 mmol), and NaOBu-*t* (0.089 g, 0.93 mmol) in toluene (1 mL) was stirred at 100 °C for 3 h. After cooling, the reaction mixture was diluted with Et_2O and washed with water and with brine. The organic layer was dried over Na_2SO_4, filtered, and concentrated under reduced pressure. The residue was dissolved in toluene (2 mL) and piperidine (0.337 mL, 3.4 mmol), $Pd_2(dba)_3$ (0.007 g, 0.0078 mmol), X-Phos (0.0148 g, 0.031 mmol), and NaOBu-*t* (0.059 g, 0.62 mmol) were added. The mixture was then stirred at 80 °C for 24 h. After cooling, the reaction mixture was dried under reduced pressure and the residue was purified by flash chromatography (SiO_2, 50 g; *n*-hexane/ethyl acetate 85:15 v/v) to give **10b** (0.045 g, 50%) as a yellow solid. Mp 139–140 °C; MS (relative intensity): *m*/*z* 291 (M^+, 100), 235 (13), 207 (17); IR (KBr) 3100, 2921, 1596, 765, 690 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.87 (d, *J*=8.4 Hz, 2H), 7.46 (t, *J*=7.6 Hz, 2H), 7.36 (d, *J*=7.2 Hz, 1H), 6.98 (s, 1H), 6.95 (s, 1H), 6.62 (s, 1H), 3.37 (br t, *J*=5.2 Hz, 4H), 2.43 (s, 3H), 1.91–1.85 (m, 4H), 1.72–1.65 (m, 2H); C NMR ($CDCl_3$): δ 155.0, 145.2, 137.8, 133.2, 130.8, 130.3, 128.7, 128.2, 124.8, 113.3, 113.1, 101.5, 51.4, 26.2, 24.6, 21.8. Anal. Calcd for $C_{20}H_{21}NO_3$: C, 82.44; H, 7.26; N, 4.81. Found: C, 81.95; H, 7.28; N, 4.78.

4.3. Characterization

*4.3.1. 2-Chloro-4-phenylphenol (**2a**).* White solid. Mp: 80–81 °C. MS (relative intensity): *m*/*z* 206 (M^+, 32), 204 (M^+, 100), 128 (22), 91 (26); IR (KBr): 3288, 3030, 1515, 1450, 1278, 761 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.59 (d, *J*=2.0 Hz, 1H), 7.55–7.53 (m, 2H), 7.47–7.43 (m, 3H), 7.37–7.34 (m, 1H), 7.11 (d, 1H), 5.58 (s, 1H); ^{13}C NMR ($CDCl_3$): δ 150.7, 139.6, 135.0, 128.9, 127.5, 127.3, 127.1, 126.7, 120.3, 116.5. Anal. Calcd for $C_{12}H_9ClO$: C, 70.43; H, 4.43; Cl, 17.32. Found: C, 70.48; H, 4.42; Cl, 17.39.

*4.3.2. 2-Bromo-4-phenylphenol (**2b**).* White solid. Mp: 94–96 °C. MS (relative intensity): *m*/*z* 250 (M^+, 92), 248 (M^+, 90), 77 (21); IR (KBr) 3293, 3039, 1450, 761 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.73 (d, *J*=2 Hz, 1H), 7.54–7.52 (m, 2H), 7.46–7.41 (m, 3H), 7.36 (t, *J*=7.2 Hz, 1H), 7.12 (d, *J*=8.4 Hz, 1H), 5.56 (s, 1H); ^{13}C NMR (CDCl3): δ 151.7, 139.4, 135.4, 130.5, 128.9, 128.0, 127.2, 126.7, 116.3, 110.7. Anal. Calcd for $C_{12}H_9BrO$: C, 57.86; H, 3.64; Br, 32.08. Found: C, 57.94; H, 3.63, Br, 32.01.

*4.3.3. 2-Chloro-6-iodo-4-methylphenol (**3c**).* Yellow oil. MS (relative intensity): *m*/*z* 270 (M^+, 32), 268 (M^+, 100), 233 (23), 142 (10), 77 (89); IR (KBr): 3488, 2921, 2360, 1471, 1160, 779 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.46 (d, *J*=1.2 Hz, 1H), 7.15 (d, *J*=1.2 Hz, 1H), 5.78 (br s, 1H), 2.26 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 163.6, 148.6, 138.0, 132.7, 130.1, 83.1, 19.9. Anal. Calcd for C_7H_6ClIO: C, 31.32; H, 2.25; Cl, 13.21. Found: C, 31.25; H, 2.24; Cl, 13.16.

*4.3.4. 2-Chloro-4-fluorophenol (**2e**).* Yellow oil. MS (relative intensity): *m*/*z* 148 (M^+, 32), 146 (M^+, 100), 111 (20), 83 (35); IR (KBr): 3430, 2927, 1114 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.1 (dd, J_1=7.6 Hz, J_2=2.8 Hz, 1H), 6.99–6.90 (m, 2H), 5.46 (s, 1H); ^{13}C NMR ($CDCl_3$): δ 156.9 (d, J_{C-F}=248 Hz), 149.0, 125.9, 118.9, 118.2 (d, J_{C-F}=22 Hz), 115.0 (d, J_{C-F}=20 Hz). Anal. Calcd for C_6H_4ClFO: C, 76.57; H, 6.43; Cl, 24.19. Found: C, 76.65; H, 6.42; Cl, 24.26.

*4.3.5. 2-Bromo-4-fluorophenol (**2f**).* White solid. Mp: 43–45 °C. MS (relative intensity): *m*/*z* 191 (M^+, 100), 189 (M^+, 100), 111 (13); IR (KBr): 3519, 3076, 1198, 768 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.23 (dd, J_1=7.6 Hz, J_2=2.8 Hz, 1H), 7.00–6.97 (m, 2H), 5.37 (s, OH); ^{13}C NMR ($CDCl_3$): δ 155.7 (d, J_{C-F}=248 Hz), 149.2, 122.1 (d, J_{C-F}=22.0 Hz), 117.4, 116.0 (d, J_{C-F}=20.0 Hz), 114.8. Anal. Calcd for C_6H_4BrFO: C, 37.73; H, 2.11; Br, 41.84. Found: C, 37.65; H, 2.12; Br, 41.91.

*4.3.6. 3-Chloro-4-hydroxyphenethyl methyl carbonate (**2g**).* White solid. Mp: 79–80 °C. MS: *m*/*z* 255 (M^+Na), 253 (M^+Na); IR (KBr):

3370, 3079, 2954, 1718, 1278, 792; ^{1}H NMR ($CDCl_3$): δ 7.18 (d, J=2 Hz 1H), 7.02 (dd, J_1=8 Hz, J_2=2 Hz, 1H), 6.93 (d, J=8 Hz, 1H), 5.99 (br s, OH), 4.30 (t, J=7.2 Hz, 2H), 3.77 (s, 3H), 2.88 (t, J=7.2 Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 155.8, 150.3, 130.4, 129.4, 128.8, 119.8, 116.4, 68.2, 54.8, 34.0. Anal. Calcd for $C_{10}H_{11}ClO_4$: C, 52.07; H, 4.81; Cl, 15.37. Found: C, 76.15; H, 4.80; Cl, 15.29.

*4.3.7. 3-Chloro-5-iodobiphenyl-4-ol (**3a**).* White solid. Mp: 90–91 °C. MS (relative intensity): m/z 332 (M^+, 16), 330 (M^+, 42), 253 (25), 205 (100), 207 (83), 73 (100); IR (KBr): 3480, 1552, 1471, 1286, 1166, 757 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.87 (d, J=2 Hz, 1H), 7.57 (d, J=2 Hz, 1H), 7.50–7.48 (m, 2H), 7.45–7.43 (m, 2H), 7.37–7.35 (m, 1H), 5.96 (br s, 1H); ^{13}C NMR ($CDCl_3$): δ 150.2, 138.1, 136.5, 136.3, 129.0, 128.3, 127.7, 126.7, 119.3, 83.9. Anal. Calcd for $C_{12}H_8ClIO$: C, 43.60; H, 2.44; Cl, 10.73. Found: C, 43.51; H, 2.43; Cl, 10.68.

*4.3.8. 3-Bromo-5-iodobiphenyl-4-ol (**3b**).* White solid. Mp: 80–81 °C. MS (relative intensity): m/z 376 (M^+, 100), 375 (M^+, 100), 168 (26); IR (KBr): 3438, 3060, 2970, 1459, 1284, 767 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.91 (d, J=2 Hz, 1H), 7.72 (d, J=2 Hz, 1H), 7.52–7.35 (m, 5H), 5.9 (s, 1H); ^{13}C NMR ($CDCl_3$): δ 150.8, 139.3, 137.98, 137.0, 131.3, 128.9, 127.7, 126.8, 109.0, 83.5. Anal. Calcd for $C_{12}H_8BrIO$: C, 38.43; H, 2.15; Br, 21.31. Found: C, 38.35; H, 2.16; Br, 21.25.

*4.3.9. 2-Bromo-6-iodo-4-methylphenol (**3d**).* Yellow solid. Mp: 52–53 °C. MS (relative intensity): m/z 311 (M^+, 100), 313 (M^+, 98), 233 (38), 185 (30), 106 (29); IR (KBr): 3400, 2940, 1467, 721 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.50 (d, J=1.2 Hz, 1H), 7.30 (d, J=1.2 Hz, 1H), 5.78 (br s, 1H), 2.26 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 149.3, 138.8, 134.1, 133.1, 108.0, 82.8, 19.7. Anal. Calcd for C_7H_6BrIO: C, 26.87; H, 1.93; Br, 25.83. Found: C, 26.80; H, 1.94; Br, 25.78.

*4.3.10. 2-Chloro-4-fluoro-6-iodophenol (**3e**).* Brown oil. MS (relative intensity): m/z 273 (M^+, 34), 271 (M^+, 100), 145 (13), 127 (38), 91 (10); IR (KBr): 3519, 3081, 1594, 1467, 1197, 784 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.40 (dd, J_1=7.6 Hz, J_2=2.8 Hz, 1H), 7.15 (dd, J_1=7.6 Hz, J_2=2.8 Hz, 1H), 5.67 (br s, 1H); ^{13}C NMR ($CDCl_3$): δ 155.8 (d, J_{C-F}=247 Hz), 147.9, 124.5 (d, 24 Hz), 118.6, 116.9 (d, 26 Hz), 82.3. Anal. Calcd for C_6H_3ClFIO: C, 26.45; H, 1.11; Cl, 13.01. Found: C, 26.54; H, 1.12, Cl, 13.07.

*4.3.11. 2-Bromo-4-fluoro-6-iodophenol (**3f**).* Brown solid. Mp: 48–50 °C. MS (relative intensity): m/z 315 (M^+, 100), 317 (M^+, 99), 110 (29); IR (KBr): 3461, 1637, 1463, 727 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.45 (dd, J_1=7.6 Hz, J_2=2.8 Hz, 1H), 7.29 (dd, J_1=7.6 Hz, J_2=2.8 Hz, 1H), 5.74 (br s, 1H); ^{13}C NMR ($CDCl_3$): δ 155.9 (d, J_{C-F}=248 Hz), 148.6, 125.1 (d, J_{C-F}=22 Hz), 119.7 (d, J_{C-F}=20 Hz), 107.5, 83.9. Anal. Calcd for C_6H_3BrFIO: C, 22.74; H, 0.95; Br, 25.21. Found: C, 22.82; H, 0.94; Br, 25.16.

*4.3.12. 3-Chloro-4-hydroxy-5-iodophenethyl methyl carbonate (**3g**).* White solid. Mp: 80–81 °C. MS: m/z 381 (MNa^+), 379 (MNa^+); IR (KBr): 3400, 3006, 2954, 1741, 1457, 1282, 956 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.51 (d, J=2.0 Hz, 1H), 7.22 (d, J=2.0 Hz, 1H), 6.01 (br s, 1H), 4.29 (t, J=7.2 Hz, 2H), 3.79 (s, 3H), 2.86 (t, J=7.2 Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 155.6, 149.7, 138.0, 123.2, 130.1, 119.0, 83.6, 67.8, 54.9, 33.5. Anal. Calcd for $C_{10}H10ClIO_4$: C, 33.69; H, 2.83; Cl, 9.94. Found: C, 33.60; H, 2.84; Cl, 9.89.

*4.3.13. 7-Chloro-2,5-diphenylbenzo[b]furan (**5a**).* White solid. Mp: 120–122 °C. MS: m/z 327 (MNa^+), 325 (MNa^+); IR (KBr): 3112, 2913, 2848, 1461, 1145, 763 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.95 (d, J=7.2 Hz, 2H), 7.68 (d, J=1.2 Hz, 1H), 7.63 (d, J=7.2 Hz, 2H), 7.54 (d, J=1.2 Hz, 1H), 7.50–7.47 (m, 4H), 7.40–7.38 (m, 2H), 7.11 (s, 1H); ^{13}C NMR ($CDCl_3$): δ 157.4, 150.1, 140.5, 137.9, 131.2, 129.8, 129.7, 128.9, 128.7, 128.5, 127.4, 125.2, 124.0, 117.9, 116.7, 101.9. Anal. Calcd for $C_{20}H_{13}ClO_4$: C, 78.82; H, 4.30; Cl, 11.63. Found: C, 78.74; H, 4.31; Cl, 11.58.

*4.3.14. 7-Bromo-2,5-diphenylbenzo[b]furan (**5b**).* White solid. Mp: 123–125 °C. MS (relative intensity): m/z 350 (M^+, 100), 348 (M^+, 100), 77 (43); IR (KBr): 3060, 2930, 1577, 1457, 1234, 1141, 761 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.94 (d, J=8.0 Hz, 2H), 7.71–7.69 (m, 2H), 7.63–7.61 (m, 2H), 7.50–7.47 (m, 4H), 7.41–7.39 (m, 2H), 7.11 (s, 1H); ^{13}C NMR ($CDCl_3$): δ 157.3, 151.5, 140.4, 138.3, 130.8, 129.8, 129.1, 128.9, 127.4, 127.3, 126.8, 125.2, 118.5, 104.0, 102.0, 29.8. Anal. Calcd for $C_{20}H_{13}BrO$: C, 68.79; H, 3.75; Br, 22.88. Found: C, 68.88; H, 3.74; Br, 22.80.

*4.3.15. 7-Chloro-5-methyl-2-phenylbenzo[b]furan (**5c**).* White solid. Mp: 123–124 °C. MS (relative intensity): m/z 244 (M^+, 32), 242 (M^+, 100), 207 (36), 77 (14); IR (KBr): 3112, 2927, 1463, 763 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.91 (d, J=8.0 Hz, 2H), 7.48–7.45 (m, 2H), 7.39 (t, J=7.2 Hz, 1H), 7.27 (s, 1H), 7.13 (s, 1H), 6.98 (s, 1H), 2.45 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 194.3, 156.9, 149.0, 133.7, 130.8, 130.0, 128.8, 125.5, 125.1, 119.3, 115.9, 101.5, 21.2. Anal. Calcd for $C_{15}H_{11}ClO$: C, 74.23; H, 4.57; Cl, 14.61. Found: C, 74.15; H, 4.58; Cl, 14.55.

*4.3.16. 7-Chloro-5-fluoro-2-phenylbenzo[b]furan (**5e**).* White solid. Mp: 74–75 °C. MS: m/z 269 (MNa^+), 267 (MNa^+); IR (KBr): 1631, 1471, 1193, 728 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.91 (d, J=8 Hz, 2H), 7.50–7.47 (m, 2H), 7.43–7.40 (m, 1H), 7.17 (dd, J_1=8.4 Hz, J_2=2.4 Hz, 1H), 7.10 (dd, J_1=8.4 Hz, J_2=2.4 Hz, 1H), 7.03 (s, 1H); ^{13}C NMR ($CDCl_3$): δ 158.0 (d, J_{C-F}=240 Hz), 157.5, 146.3, 129.8, 128.5, 128.4, 127, 124.3, 115.8 (d, J_{C-F}=12 Hz), 111.5 (d, J_{C-F}=29 Hz), 104.1 (d, J_{C-F}=25 Hz), 100.9. Anal. Calcd for $C_{14}H_8ClFO$: C, 68.17; H, 3.27; Cl, 14.37. Found: C, 68.09; H, 3.28; Cl, 14.30.

*4.3.17. 7-Bromo-5-fluoro-2-phenylbenzo[b]furan (**5f**).* White solid. Mp: 59–60 °C. MS: m/z 313 (MNa^+), 311 (MNa^+); IR (KBr): 3087, 2927, 1631, 1577, 1469, 1193, 755 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.90 (d, J=7.2 Hz, 2H), 7.50–7.47 (m, 2H), 7.42–7.40 (m, 1H), 7.23 (m, 2H), 7.06 (s, 1H); ^{13}C NMR ($CDCl_3$): δ 162.0 (d, J_{C-F}=240 Hz), 161.3, 151.6, 133.2, 132.4, 132.2, 131.8, 128.1, 118.1 (d, J_{C-F}=29 Hz), 108.6 (d, J_{C-F}=25 Hz), 106.4 (d, J_{C-F}=12 Hz), 104.8 (d, J_{C-F}=4 Hz). Anal. Calcd for $C_{14}H_8BrO$: C, 57.76; H, 2.77; Br, 27.45. Found: C, 57.82; H, 2.78; Br, 27.51.

*4.3.18. 7-Chloro-2-pentyl-5-phenylbenzo[b]furan (**5g**).* White solid. Mp: 58–59 °C. MS: m/z 321 (MNa^+), 319 (MNa^+); IR (KBr): 2929, 1597, 1572, 1462, 1444 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.63–7.59 (m, 3H), 7.50–7.46 (m, 3H), 7.41–7.37 (m, 1H), 6.49 (s, 1H), 2.85 (t, J=7.6 Hz, 2H), 1.85–1.79 (m, 2H), 1.47–1.43 (m, 4H), 0.97 (t, J=8 Hz, 3H); ^{13}C NMR ($CDCl_3$): δ 161.6, 149.9, 140.7, 137.4, 131.1, 128.8, 127.4, 127.2, 122.8, 117.3, 116.2, 102.7, 31.4, 28.5, 27.3, 22.4, 14.0. Anal. Calcd for $C_{19}H_{19}ClO$: C, 76.37; H, 6.41; Cl, 11.86. Found: C, 76.45; H, 6.42; Cl, 11.80.

*4.3.19. 7-Chloro-5-methyl-2-phenethylbenzo[b]furan (**5h**).* Yellow oil. MS: m/z 293 (MNa^+), 291 (MNa^+); IR (KBr): 2924, 1601, 1473, 1456 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.34–7.24 (m, 5H), 7.16 (s, 1H), 7.07 (s, 1H), 6.33 (s, 1H), 3.13–3.11 (m, 4H), 2.42 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 159.6, 148.7, 140.7, 133.2, 130.5, 128.5, 128.4, 126.2, 124.5, 118.9, 115.5, 102.8, 33.8, 30.2, 21.1. Anal. Calcd for $C_{17}H_{15}ClO$: C, 75.41; H, 5.58; Cl, 13.09. Found: C, 75.50; H, 5.59; Cl, 13.02.

*4.3.20. 2-(7-Chloro-2-phenylbenzo[b]furan-5-yl)ethyl acetate (**5i**).* White solid. Mp: 103–104 °C. MS (relative intensity): m/z 256 (M^+, 27), 254 (100), 78 (26); IR (KBr): 3073, 2954, 1747, 1467, 1265, 734 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.90 (d, J=7.2 Hz, 2H), 7.48 (m, 2H), 7.4 (t, J=7.6 Hz, 1H), 7.34 (s, 1H), 7.18 (s, 1H), 7.00 (s, 1H), 4.40 (t, J=6.8 Hz, 2H), 3.8 (s, 3H), 3.06 (t, J=6.8 Hz, 2H); ^{13}C NMR

($CDCl_3$): δ 157.2, 155.7, 149.7, 139.7, 133.3, 131.0, 129.8, 129.0, 128.8, 125.1, 119.5, 116.4, 101.5, 68.4, 54.8, 34.9. Anal. Calcd for $C_{18}H_{15}ClO_3$: C, 68.68; H, 4.80; Cl, 11.26. Found: C, 68.59; H, 4.79; Cl, 11.19.

*4.3.21. 7-Bromo-2-(4-methoxyphenyl)-5-methylbenzo[b]furan (**5j**).* White solid. Mp: 133–135 °C. MS: *m/z* 339 (MNa^+), 337 (MNa^+); IR (KBr): 2960, 1610, 1577, 1504, 1464, 1254, 1036 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.84 (d, *J*=8.8 Hz, 2H), 7.27–7.25 (m, 2H), 7.01 (d, *J*=8.8 Hz, 2H), 6.87 (s, 1H), 3.89 (s, 3H), 2.44 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 160.3, 156.9, 150.2, 134.0, 129.8, 127.7, 122.8, 119.7, 115.7, 114.3, 103.1, 100.0, 55.4, 21.1. Anal. Calcd for $C_{16}H_{13}BrO_2$: C, 60.59; H, 4.13; Br, 25.19. Found: C, 60.50; H, 4.14; Br, 25.12.

*4.3.22. 1-(7-Bromo-5-phenylbenzo[b]furan-2-yl)-1-phenylethanol (**5k**).* White-yellow. Mp: 45–47 °C. MS: *m/z* 417 (MNa^+), 415 (MNa^+); IR (KBr): 3408, 2925, 1572, 1493, 1460, 1444 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.69–7.66 (m, 2H), 7.62–7.36 (m, 10H), 6.73 (s, 1H), 3.65 (br s, 1H), 2.06 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 163.1, 151.6, 144.7, 140.4, 138.3, 129.8, 128.9, 128.4, 127.8, 127.5, 126.9, 125.4, 118.9, 110.1, 104.2, 104.1, 73.5, 29.1. Anal. Calcd for $C_{22}H_{17}BrO_2$: C, 67.19; H, 4.36; Br, 20.32. Found: C, 67.10; H, 4.37; Br, 20.25.

*4.3.23. Methyl 3-(2,5-diphenylbenzo[b]furan-7-yl)benzoate (**6aa**).* White solid. Mp: 143–145 °C. MS: *m/z* 427 (MNa^+); IR (KBr): 3112, 2954, 1712, 1297, 1218, 756 cm^{-1}; ^1H NMR ($CDCl_3$): δ 8.74 (s, 1H), 8.25 (d, *J*=7.6 Hz, 1H), 8.16 (d, *J*=8.0 Hz, 1H), 7.93–7.91 (m, 2H), 7.80 (d, *J*=2 Hz, 1H), 7.75–7.65 (m, 4H), 7.54–7.48 (m, 4H), 7.43–7.40 (m, 2H), 7.153 (s, 1H), 4.02 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 167.1, 156.8, 151.6, 141.5, 137.4, 136.7, 132.9, 130.7, 130.2, 129.7, 128.9, 128.83, 128.81, 128.77, 127.5, 127.1, 125.0, 124.1, 124.1, 123.5, 123.4, 119.1, 101.6, 52.2. Anal. Calcd for $C_{28}H_{20}O_3$: C, 83.15; H, 4.98. Found: C, 83.23; H, 4.99.

*4.3.24. 7-(4-Methoxyphenyl)-2,5-diphenylbenzo[b]furan (**6ab**).* White solid. Mp: 137–138 °C. MS: *m/z* 399 (MNa^+); IR (KBr): 3006, 2821, 1606, 1513, 1461, 1243, 1180, 752 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.97 (d, *J*=8 Hz, 2H), 7.93–7.91 (m, 2H), 7.74–7.68 (m, 4H), 7.53–7.48 (m, 4H), 7.42–7.37 (m, 2H), 7.16–7.13 (m, 3H), 3.95 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 159.4, 156.6, 151.6, 141.8, 137.3, 130.6, 130.4, 128.9, 128,82, 128.79, 128.6, 127.5, 126.9, 125.0, 123.2, 117.98, 114.2, 101.6, 55.4. Anal. Calcd for $C_{27}H_{20}O_2$: C, 86.14; H, 5.36. Found: C, 86.23; H, 5.37.

*4.3.25. 7-(Biphenyl-4-yl)-2,5-diphenylbenzo[b]furan (**6ac**).* White solid. Mp: 181–183 °C. MS: *m/z* 445 (MNa^+); IR (KBr): 3033, 2940, 2861, 1604. 1446, 1218, 762 cm^{-1}; ^1H NMR ($CDCl_3$): δ 8.4 (d, *J*=8.0 Hz, 2H), 7.95 (d, *J*=7.2 Hz, 2H), 7.85 (d, *J*=8.0 Hz, 2H), 7.80–7.74 (m, 8H), 7.54–7.42 (m, 9H), 7.17 (s, 1H); ^{13}C NMR ($CDCl_3$): δ 159.7, 151.8, 141.7, 140.8, 140.6, 137.4, 135.5, 130.7, 130.3, 129.0, 128.89, 128.86, 128.85, 128.76, 128.5, 127.5, 127.4, 127.2, 127.1, 125.0, 123.9, 123.6, 118.7, 101.7. Anal. Calcd for $C_{32}H_{22}O$: C, 90.97; H, 5.25. Found: C, 90.89; H, 5.26.

*4.3.26. 7-(4-Fluoro-3-methylphenyl)-2,5-diphenylbenzo[b]furan (**6ad**).* White solid. Mp: 126–127 °C. MS (relative intensity): *m/z* 378 (M^+, 100), 364 (19), 301 (23), 108 (16), 77 (17); IR (KBr): 3046, 2927, 1504, 1465, 1234, 756 cm^{-1}; ^1H NMR ($CDCl_3$): δ 9.32–9.07 (m, 8H), 8.93–8.83 (m, 6H), 8.69–8.63 (m, 1H), 8.55 (s, 1H), 3.88 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 160.4 (d, J_{C-F}=245 Hz), 156.6, 151.5, 141.6, 137.2, 132.2, 131.7, 130.6, 130.3, 128.8, 128.7, 127.7, 127.6, 127.5, 127.0, 125.1, 125.0, 124.5, 123.4, 118.4, 115.2 (d, J_{C-F}=22 Hz), 101.7, 14.8. Anal. Calcd for $C_{27}H_{19}FO$: C, 85.69; H, 5.06. Found: C, 85.78; H, 5.07.

*4.3.27. 2,5-Diphenyl-7-(thiophen-3-yl)benzo[b]furan (**6ae**).* White solid. Mp: 136–137 °C. MS: *m/z* 375 (MNa^+); IR (KBr): 3033, 2927, 1600, 1467, 1228, 786, 762 cm^{-1}; ^1H NMR ($CDCl_3$): δ 8.15 (dd, J_1=2.8 Hz, J_2=1.2 Hz, 1H), 7.96–7.94 (m, 2H), 7.86 (dd, J_1=4.0 Hz, J_2=1.2 Hz, 1H), 7.81 (d, *J*=1.6 Hz, 1H), 7.73–7.71 (m, 3H), 7.55–7.51 (m, 5H), 7.45–7.41 (m, 2H), 7.12 (s, 1H); ^{13}C NMR ($CDCl_3$): δ 156.6, 151.3, 141.7, 137.2, 136.8, 130.6, 130.3, 128.9, 128.8, 128.7, 127.6, 127.1, 125.6, 125.1, 123.3, 122.1, 120.1, 118.3, 101.8. Anal. Calcd for $C_{24}H_{16}OS$: C, 81.79; H, 4.58. Found: C, 81.70; H, 4.59.

*4.3.28. 7-(4-Methoxyphenyl)-5-methyl-2-phenylbenzo[b]furan (**6cb**).* White solid. Mp: 127–128 °C. MS: *m/z* 337 (MNa^+); IR (KBr): 3060, 2941, 1608, 1513, 1384, 1245, 833, 746 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.95–7.87 (m, 4H), 7.50–7.35 (m, 5H), 7.13–7.04 (m, 3H), 3.94 (s, 3H), 2.53 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 159.2, 156.0, 150.4, 132.8, 130.6, 130.1, 129.6, 129.5, 129.1, 128.7, 128.4, 124.9, 124.7, 119.4, 114.1, 101.2, 55.3, 21.4. Anal. Calcd for $C_{22}H_{18}O_2$: C, 84.05; H, 5.77. Found: C, 84.14; H, 5.78.

*4.3.29. 7-(4-Methoxyphenyl)-5-methyl-2-phenylbenzo[b]furan (**6cb**).* White solid. Mp: 127–128 °C. MS: *m/z* 337 (MNa^+); IR (KBr): 3060, 2941, 1608, 1513, 1384, 1245, 833, 746 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.95–7.87 (m, 4H), 7.50–7.35 (m, 5H), 7.13–7.04 (m, 3H), 3.94 (s, 3H), 2.53 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 159.2, 156.0, 150.4, 132.8, 130.6, 130.1, 129.6, 129.5, 129.1, 128.7, 128.4, 124.9, 124.7, 119.4, 114.1, 101.2, 55.3, 21.4. Anal. Calcd for $C_{22}H_{18}O_2$: C, 84.05; H, 5.77. Found: C, 84.14; H, 5.78.

*4.3.30. 7-(4-Fluoro-3-methylphenyl)-5-methyl-2-phenylbenzo[b]furan (**6cd**).* White solid. Mp: 81–82 °C. MS: *m/z* 334 (MNa^+); IR (KBr): 3033, 2927, 1617, 1508, 1238, 746 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.89 (d, *J*=7.6 Hz, 2H), 7.79 (d, *J*=6.8 Hz, 2H), 7.51–7.48 (m, 2H), 7.42–7.37 (m, 2H), 7.27–7.21 (m, 2H), 7.03 (s, 1H), 2.55 (s, 3H), 2.48 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 161.3 (d, J_{C-F}=245 Hz), 156.1, 150.4, 132.9, 132.4, 131.6, 130.5, 130.1, 128.8, 128.5, 127.6, 127.5, 125.0, 124.9, 123.9, 119.9, 115.1 (d, J_{C-F}=22 Hz), 101.3, 21.4, 14.8. Anal. Calcd for $C_{22}H_{17}FO$: C, 83.52; H, 5.42. Found: C, 83.45; H, 5.41.

*4.3.31. 5-Methyl-2-phenyl-7-(thiophen-3-yl)benzo[b]furan (**6ce**).* White solid. Mp: 107.5–108.5 °C. MS: *m/z* 313 (MNa^+); IR (KBr): 2921, 1606, 1384, 1203, 783, 762 cm^{-1}; ^1H NMR ($CDCl_3$): δ 8.11–8.10 (m, 1H), 7.94–7.92 (m, 2H), 7.82–7.81 (m, 1H), 7.54–7.34 (m, 6H), 7.03 (s, 1H), 2.54 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 156.1, 150.2, 148.2, 132.8, 130.5, 128.8, 128.5, 127.0, 125.4, 124.9, 123.9, 122.9, 119.7, 119.4, 101.4, 21.5. Anal. Calcd for $C_{19}H_{14}OS$: C, 78.59; H, 4.86. Found: C, 78.50; H, 4.87.

*4.3.32. Methyl 3-(5-fluoro-2-phenylbenzo[b]furan-7-yl)benzoate (**6ea**).* White solid. Mp: 111–112 °C. MS: *m/z* 369 (MNa^+); IR (KBr): 3060, 2940, 1730, 1606, 1473, 1448, 1400, 1294, 752 cm^{-1}; ^1H NMR ($CDCl_3$): δ 8.68 (s, 1H), 8.15 (t, *J*=7.6 Hz, 2H), 7.87 (d, *J*=7.2 Hz, 2H), 7.65 (t, *J*=7.6 Hz, 1H), 7.50–7.38 (m, 3H), 7.27–7.25 (m, 2H), 7.07 (s, 1H), 4.02 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 166.9, 159.5 (d, J_{C-F}=237 Hz), 157.8, 148.2, 138.8, 136.8, 135.7, 132.7, 129.9, 129.2, 129.0, 128.9, 128.8, 128.5, 125.0, 111.1 (d, J_{C-F}=26 Hz), 105.9 (d, J_{C-F}=25 Hz), 101.6 (d, J_{C-F}=4 Hz), 52.3. Anal. Calcd for $C_{22}H_{15}FO$: C, 76.29; H, 4.37. Found: C, 76.20; H, 4.38.

*4.3.33. 5-Fluoro-7-(4-methoxyphenyl)-2-phenylbenzo[b]furan (**6eb**).* White solid. Mp: 90–92 °C. MS: *m/z* 341 (MNa^+); IR (KBr): 2923, 1600, 1517, 1469, 1390, 1290, 1180, 761 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.93–7.86 (m, J_1=4 Hz), 7.51–7.39 (m, 3H), 7.21–7.12 (m, 4H), 7.04 (s, 1H), 3.94 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 160.0 (d, J_{C-F}=237 Hz), 159.7, 157.5, 148.2, 130.7, 130.1, 129.7, 128.8, 128.0, 127.5, 127.4, 125.1, 114.3, 110.5 (d, J_{C-F}=26 Hz), 104.6 (d, J_{C-F}=25 Hz), 101.6 (d,

J_{C-F}=4 Hz), 55.4. Anal. Calcd for $C_{21}H_{15}FO_2$: C, 79.23; H, 4.75. Found: C, 79.15; H, 4.76.

*4.3.34. 7-(Biphenyl-4-yl)-5-fluoro-2-phenylbenzo[b]furan (**6ec**).* White solid. Mp: 173–175 °C. MS: *m*/*z* 387 (MNa^+); IR (KBr): 3033, 2930, 1604, 1471, 1390, 761 cm^{-1}; ^1H NMR ($CDCl_3$): δ 8.05 (d, *J*=8.4 Hz, 2H), 7.90 (d, *J*=7.2 Hz, 2H), 7.82 (d, *J*=8.4 Hz, 2H), 7.74 (d, *J*=7.2 Hz, 2H), 7.55–7.41 (m, 6H), 7.29–7.24 (m, 2H), 7.09 (s, 1H); ^{13}C NMR ($CDCl_3$): δ 159.2 (d, J_{C-F}=237 Hz), 157.7, 148.6, 148.4, 141.0, 140.6, 134.5, 130.9, 130.7, 130.0, 129.0, 128.9, 128.8, 127.6, 127.5, 126.1, 125.1, 110.9 (d, J_{C-F}=27 Hz), 105.3 (d, J_{C-F}=25 Hz), 101.5 (d, J_{C-F}=4 Hz). Anal. Calcd for $C_{26}H_{17}FO$: C, 85.69; H, 4.70. Found: C, 85.60; H, 4.71.

*4.3.35. 5-Fluoro-7-(4-fluoro-3-methylphenyl)-2-phenylbenzo[b]furan (**6ed**).* White solid. Mp: 106–107 °C. MS: *m*/*z* 343 (MNa^+); IR (KBr): 3079, 2923, 1604, 1469, 1234, 759 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.85 (d, *J*=7.2 Hz, 2H), 7.74 (d, *J*=6.8 Hz, 2H), 7.50 (t, *J*=7.2 Hz, 2H), 7.42 (t, *J*=7.2 Hz, 1H), 7.24–7.14 (m, 3H), 7.02 (s, 1H), 2.45 (d, *J*=1.6 Hz, 3H); ^{13}C NMR ($CDCl_3$): δ 161.3 (d, J_{C-F}=245 Hz), 159.6 (d, J_{C-F}=237 Hz), 158.4 (d, J_{C-F}=237 Hz), 157.6, 148.2, 131.7, 131.6, 130.8, 130.7, 130.0, 129.0, 128.9, 127.5, 125.0, 115.3 (d, J_{C-F}=22 Hz), 110.9 (d, J_{C-F}=27 Hz), 105.2 (d, J_{C-F}=25 Hz), 101.5 (d, J_{C-F}=4 Hz), 14.7 (d, J_{C-F}=3 Hz). Anal. Calcd for $C_{21}H_{14}F_2O$: C, 78.74; H, 4.41. Found: C, 78.67; H, 4.40.

*4.3.36. 5-Fluoro-2-phenyl-7-(thiophen-3-yl)benzo[b]furan (**6ee**).* White solid. Mp: 114–115 °C. MS: *m*/*z* 317 (MNa^+); IR (KBr): 3093, 2921, 1604, 1471, 1384, 1137, 761 cm^{-1}; ^1H NMR ($CDCl_3$): δ 8.10–8.09 (m, 1H), 7.89 (d, *J*=7.6 Hz, 2H), 7.74 (dd, *J*=5.2, 1.2 Hz, 1H), 7.53–7.49 (m, 3H), 7.44–7.40 (m, 1H), 7.30–7.27 (m, 1H), 7.18 (dd, *J*=8.0 Hz, *J*=2.4 Hz, 1H), 7.03 (s, 1H); ^{13}C NMR ($CDCl_3$): δ 159.5 (d, J_{C-F}=237 Hz), 157.6, 147.9, 135.9, 130.6, 130.0, 128.9, 126.8, 125.5, 123.9, 120.8, 120.7, 110.0 (d, J_{C-F}=27 Hz), 105.0 (d, J_{C-F}=25 Hz), 101.7 (d, J_{C-F}=4 Hz). Anal. Calcd for $C_{18}H_{11}FOS$: C, 73.45; H, 3.77. Found: C, 73.38; H, 3.78.

*4.3.37. 3-(5-(2-(Methoxycarbonyloxy)ethyl)-2-phenylbenzo[b]furan-7-yl)benzoate (**6ia**).* White solid. Mp: 103–104 °C. MS: *m*/*z* 453 (MNa^+); IR (KBr): 2954, 1737, 1714, 1604, 1440, 1384, 1274, 759 cm^{-1}; ^1H NMR ($CDCl_3$): δ 8.68 (s, 1H), 8.19 (d, *J*=7.6 Hz, 1H), 8.13 (d, *J*=7.6 Hz, 1H), 7.88 (d, *J*=7.2 Hz, 2H), 7.64 (t, *J*=8 Hz, 1H), 7.49–7.45 (m, 3H), 7.40–7.36 (m, 2H), 7.06 (s, 1H), 4.47 (t, *J*=7.2 Hz, 2H), 4.02 (s, 3H), 3.81 (s, 3H), 3.16 (t, *J*=7.2 Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 172.8, 167.1, 156.6, 155.8, 151.1, 136.6, 132.9, 132.6, 130.7, 130.5, 130.2, 129.7, 128.8, 128.7, 125.0, 124.9, 124.6, 123.9, 120.7, 101.3, 68.7, 54.8, 52.2, 35.2. Anal. Calcd for $C_{26}H_{22}O_6$: C, 72.55; H, 5.15. Found: C, 72.48; H, 5.16.

*4.3.38. 2-(7-(4-Methoxyphenyl)-2-phenylbenzo[b]furan-5-yl)ethylmethyl carbonate (**6ib**).* Brown solid. Mp: 112–113 °C. MS: *m*/*z* 425 (MNa^+); IR (KBr): 2950, 1749, 1270, 748 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.93–7.88 (m, 4H), 7.47 (t, *J*=7.6 Hz, 2H), 7.41–7.36 (m, 2H), 7.29 (d, *J*=7.6 Hz, 1H), 7.11 (d, *J*=8.8 Hz, 2H), 7.06 (s, 1H), 4.46 (t, *J*=7.2 Hz, 2H), 3.93 (s, 3H), 3.81 (s, 3H), 3.15 (t, *J*=7.2 Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 159.3, 156.3, 155.8, 151.1, 132.4, 130.4, 130.3, 129.7, 128.8, 128.78, 128.6, 125.0, 124.9, 124.4, 119.5, 114.2, 101.3, 68.9, 55.4, 54.7, 35.2. Anal. Calcd for $C_{25}H_{22}O_5$: C, 74.61; H, 5.51. Found: C, 74.53; H, 5.52.

*4.3.39. 2-(7-(Biphenyl-4-yl)-2-phenylbenzo[b]furan-5-yl)ethylmethyl carbonate (**6ic**).* White solid. Mp: 200 °C dec. MS: *m*/*z* 471 (MNa^+); IR (KBr): 2956, 1741, 1604, 1439, 1269, 1024 cm^{-1}; ^1H NMR ($CDCl_3$): δ 8.07 (d, *J*=7.6 Hz, 2H), 7.91 (d, *J*=7.2 Hz, 2H), 7.82 (d, *J*=8.0 Hz, 2H), 7.75 (d, *J*=7.2 Hz, 2H), 7.55–7.37 (m, 8H), 7.08 (s, 1H), 4.49 (t, *J*=7.2 Hz, 2H), 3.82 (s, 3H), 3.18 (t, *J*=7.2 Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 156.5, 155.8, 151.2, 140.8, 140.5, 135.4, 132.5, 130.5, 130.4, 124.9, 128.9, 128.86, 128.82, 127.5, 127.4, 127.1, 125.0, 124.8, 124.7, 120.3, 101.3, 68.8, 54.8, 35.2. Anal. Calcd for $C_{30}H_{24}O_4$: C, 80.34; H, 5.39. Found: C, 80.43; H, 5.40.

*4.3.40. 2-(7-(4-Fluoro-3-methylphenyl)-2-phenylbenzo[b]furan-5-yl)ethylmethyl carbonate (**6id**).* Brown wax. MS: *m*/*z* 427 (MNa^+); IR (KBr): 2950, 1749, 1617, 1508, 1444, 1272, 760 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.87 (d, *J*=7.2 Hz, 2H), 7.76 (d, *J*=6.4 Hz, 2H), 7.48 (t, *J*=7.6 Hz, 2H), 7.43–7.37 (m, 2H), 7.29 (s, 1H), 7.21–7.19 (m, 1H), 7.05 (s, 1H), 4.47 (t, *J*=7.2 Hz, 2H), 3.82 (s, 3H), 3.16 (t, *J*=7.2 Hz, 2H), 2.45 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 160.4 (d, J_{C-F}=237 Hz), 156.5, 155.8, 151.0, 132.5, 131.7, 131.6, 130.4, 130.3, 128.8, 128.7, 127.6, 127.5, 124.9, 124.7, 124.4, 120.1, 115.2 (d, J_{C-F}=22 Hz), 101.3, 68.8, 54.8, 35.2, 14.8. Anal. Calcd for $C_{25}H_{21}FO_4$: C, 74.24; H, 5.23. Found: C, 74.33; H, 5.24.

*4.3.41. Methyl-2-(2-phenyl-7-(thiophen-3-yl)benzo[b]furan-5-yl) ethyl carbonate (**6ie**).* Brown solid. Mp: 93–94 °C. MS: *m*/*z* 401 (MNa^+); IR (KBr): 3099, 2940, 1745, 1444, 1263, 786 cm^{-1}; ^1H NMR ($CDCl_3$): δ 8.08 (s, 1H), 7.91 (d, *J*=8.4 Hz, 2H), 7.79 (d, *J*=4.8 Hz, 1H), 7.51–7.48 (m, 3H), 7.44–7.38 (m, 3H), 7.04 (s, 1H), 4.46 (t, *J*=7.2 Hz, 2H), 3.81 (s, 3H), 3.14 (t, *J*=7.2 Hz, 2H); ^{13}C NMR ($CDCl_3$): δ 156.4, 155.8, 150.7, 136.7, 132.6, 130.4, 130.3, 128.9, 128.7, 127.0, 125.3, 125.1, 125.0, 123.6, 123.2, 119.9, 101.4, 68.8, 54.8, 35.2. Anal. Calcd for $C_{22}H_{18}O_4S$: C, 69.82; H, 4.79. Found: C, 69.74; H, 4.80.

*4.3.42. Methyl 4-(2-pentyl-5-phenylbenzo[b]furan-7-yl)benzoate (**6ga**).* White solid. Mp: 83–85 °C. MS: *m*/*z* 421 (MNa^+); IR (KBr): 2935, 1712, 1606, 1564, 1514, 1466, 1281, 1020 cm^{-1}; ^1H NMR ($CDCl_3$): δ 8.26 (d, *J*=8.4 Hz, 2H), 8.07 (d, *J*=8.4 Hz, 2H), 7.74–7.70 (m, 4H), 7.52 (t, *J*=7.6 Hz, 2H), 7.44–7.40 (m, 1H), 6.53 (s, 1H), 4.02 (s, 3H), 2.88–2.84 (m, 2H), 1.86–1.82 (m, 2H), 1.48–1.46 (m, 4H), 1.01 (t, *J*=6.8 Hz, 3H); ^{13}C NMR ($CDCl_3$): δ 167.0, 161.0, 151.5, 141.7, 141.4, 136.9, 130.7, 129.9, 129.1, 128.8, 128.6, 127.5, 127.0, 123.7, 122.5, 119.0, 102.3, 52.1, 31.4, 28.5, 27.4, 22.5, 14.1. Anal. Calcd for $C_{27}H_{26}O_3$: C, 81.38; H, 6.58. Found: C, 81.47; H, 6.59.

*4.3.43. 7-(4-fluoro-3-methylphenyl)-2-pentyl-5-phenylbenzo[b]furan (**6gb**).* Yellow oil. MS: *m*/*z* 395 (MNa^+); IR (KBr): 2929, 1603, 1504, 1466, 1444 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.82 (m, 2H), 7.75 (m, 3H), 7.66 (s, 1H), 7.56 (t, *J*=7.6 Hz, 2H), 7.45–7.43 (m, 1H), 7.25–7.22 (m, 1H), 6.56 (s, 1H), 2.91 (t, *J*=7.2 Hz, 2H), 2.50 (s, 3H), 1.91–1.85 (m, 2H), 1.53 (m, 4H), 1.06 (t, *J*=6.8 Hz, 3H); ^{13}C NMR ($CDCl_3$): δ 161.5 (d, J_{C-F}=244 Hz), 160.8, 151.5, 142.0, 136.8, 132.5, 130.5, 128.8, 127.8, 127.7, 127.6, 126.9, 125.1, 124.2, 122.4, 118.0, 115.1 (d, J_{C-F}=22 Hz), 102.3, 31.5, 28.5, 27.5, 22.5, 14.1. Anal. Calcd for $C_{26}H_{25}FO$: C, 83.84; H, 6.77. Found: C, 83.90; H, 6.78.

*4.3.44. 7-(Biphenyl-4-yl)-2-pentyl-5-phenylbenzo[b]furan (**6gc**).* White solid. Mp: 100–102 °C. MS: *m*/*z* 439 (MNa^+); IR (KBr): 2931, 1606, 1577, 1518, 1464 cm^{-1}; ^1H NMR ($CDCl_3$): δ 8.15 (d, *J*=8.4 Hz, 2H), 7.89 (d, *J*=8.4 Hz, 2H), 7.82–7.78 (m, 6H), 7.59–7.55 (m, 4H), 7.50–7.49 (m, 2H), 6.60 (s, 1H), 2.95 (t, *J*=7.2 Hz, 2H), 1.95–1.89 (m, 2H), 1.55 (m, 4H), 1.08 (t, *J*=7.2 Hz, 3H); ^{13}C NMR ($CDCl_3$): δ 160.9, 151.7, 142.0, 141.0, 140.5, 136.9, 135.8, 130.6, 129.1, 129.0, 128.9, 127.6, 127.5, 127.4, 127.2, 127.0, 124.5, 122.5, 118.2, 102.3, 31.5, 28.6, 27.5, 22.6, 14.2. Anal. Calcd for $C_{31}H_{28}O$, 89.38; H, 6.78. Found: C, 89.30; H, 6.79.

*4.3.45. 2-Pentyl-5-phenyl-7-(thiophen-3-yl)benzo[b]furan (**6gd**).* White solid. Mp: 77–79 °C. MS: *m*/*z* 369 (MNa^+); IR (KBr): 2925, 1606, 1574, 1466 cm^{-1}; ^1H NMR ($CDCl_3$): δ 8.12 (s, 1H), 7.85–7.43 (m, 9H), 6.54 (s, 1H), 2.92 (t, *J*=7.2 Hz, 2H), 1.92–1.86 (m, 2H), 1.51 (m, 4H), 1.04 (t, *J*=7.2 Hz, 3H); ^{13}C NMR ($CDCl_3$): δ 160.6, 151.2, 142.0, 137.1, 136.7, 130.5, 128.8, 127.6, 127.1, 126.9, 125.5, 123.2,

121.3, 119.7, 117.8, 102.3, 31.5, 28.6, 27.5, 22.5, 14.1. Anal. Calcd for $C_{23}H_{22}OS$: C, 79.73; H, 6.40. Found: C, 79.64; H, 6.41.

*4.3.46. 2-Pentyl-5,7-diphenylbenzo[b]furan (**6ge**).* Yellow oil. MS: *m*/*z* 363 (MNa^+); IR (KBr): 2927, 1603, 1577, 1498, 1464 cm^{-1}; ^1H NMR ($CDCl_3$): δ 8.05–8.02 (m, 2H), 7.80–7.75 (m, 4H), 7.66–7.44 (m, 6H), 6.60 (s, 1H), 2.92 (t, *J*=7.2 Hz, 2H), 1.92–1.87 (m, 2H), 1.55–1.49 (m, 4H), 1.06 (t, *J*=7.2 Hz, 3H); ^{13}C NMR ($CDCl_3$): δ 160.8, 151.6, 142.0, 136.9, 136.8, 130.6, 128.9, 128.8, 128.7, 127.7, 127.6, 126.9, 125.0, 122.6, 118.1, 102.3, 31.5, 28.6, 27.5, 22.5, 14.1. Anal. Calcd for $C_{25}H_{24}O$: C, 88.20; H, 7.11. Found: C, 88.29; H, 7.12.

*4.3.47. Methyl 4-(5-methyl-2-phenethylbenzo[b]furan-7-yl)benzoate (**6ha**).* White solid. Mp: 114–115 °C. MS: *m*/*z* 393 (MNa^+); IR (KBr): 2947, 1711, 1604, 1564, 1496, 1437, 1277, 1018 cm^{-1}; ^1H NMR ($CDCl_3$): δ 8.22 (d, *J*=8.4 Hz, 2H), 7.97 (d, *J*=8.4 Hz, 2H), 7.38–7.27 (m, 7H), 6.42 (s, 1H), 4.02 (s, 3H), 3.19–3.12 (m, 4H), 2.54 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 167.1, 159.1, 150.3, 141.5, 140.8, 132.5, 130.1, 129.9, 128.9, 128.6, 128.5, 128.4, 126.3, 124.2, 123.2, 120.6, 102.4, 52.1, 34.1, 30.3, 21.4. Anal. Calcd for $C_{25}H_{22}O_3$: C, 81.06; H, 5.99. Found: C, 81.15; H, 5.98.

*4.3.48. 7-(4-Fluoro-3-methylphenyl)-5-methyl-2-phenethylbenzo[b]furan (**6hb**).* Yellow oil. MS: *m*/*z* 367 (MNa^+); IR (KBr): 3046, 2924, 1603, 1508, 1234, 756 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.72–7.66 (m, 2H), 7.38–7.11 (m, 8H), 6.42 (s, 1H), 3.17–3.14 (m, 4H), 2.54 (s, 3H), 2.45 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 160.9 (d, J_{C-F}=244 Hz), 158.9, 150.2, 140.9, 132.4, 131.7, 131.6, 129.9, 128.5, 127.7, 126.2, 124.9, 124.8, 124.0, 123.6, 119.5, 115.0 (d, J_{C-F}=22 Hz), 102.4, 34.1, 30.3, 21.4, 14.8. Anal. Calcd for $C_{24}H_{21}FO$: C, 83.69; H, 6.15. Found: C, 83.78; H, 6.16.

*4.3.49. 7-(Biphenyl-4-yl)-5-methyl-2-phenethylbenzo[b]furan (**6hc**).* White solid. Mp: 85–87 °C. MS: *m*/*z* 411 (MNa^+); IR (KBr): 2924, 1603, 1577, 1491, 1452 cm^{-1}; ^1H NMR ($CDCl_3$): δ 8.03 (d, *J*=8 Hz, 2H), 7.84–7.78 (m, 4H), 7.59–7.32 (m, 10H), 6.46 (s, 1H), 3.21 (s, 4H), 2.60 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 158.9, 150.5, 141.0, 140.3, 135.9, 132.5, 130.1, 129.0, 128.9, 128.6, 128.5, 127.4, 127.3, 127.2, 126.3, 124.1, 124.0, 119.7, 111.1, 102.4, 34.2, 30.4, 21.5. Anal. Calcd for $C_{29}H_{24}O$: C, 89.66; H, 6.23. Found: C, 89.75; H, 6.24.

*4.3.50. 5-Methyl-2-phenethyl-7-(thiophen-3-yl)benzo[b]furan (**6hd**).* White solid. Mp: 68–69. MS: *m*/*z* 341 (MNa^+); IR (KBr): 3101, 2925, 1601, 1495, 1450 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.93 (s, 1H), 7.72 (d, *J*=4.8 Hz, 1H), 7.48–7.46 (m, 1H), 7.38–7.26 (m, 7H), 6.40 (s, 1H), 3.23–3.15 (m, 4H), 2.52 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 158.7, 150.0, 141.0, 137.1, 132.2, 129.8, 128.5, 128.4, 127.0, 126.3, 125.3, 122.9, 119.3, 119.1, 102.4, 34.2, 30.3, 21.4. Anal. Calcd for $C_{21}H_{18}OS$: C, 79.21; H, 5.70. Found: C, 79.15; H, 5.71.

*4.3.51. 5-Methyl-2-phenethyl-7-phenylbenzo[b]furan (**6he**).* Yellow oil. MS: *m*/*z* 335 (MNa^+); IR (KBr): 2922, 1603, 1577, 1496, 1471 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.96 (d, *J*=7.2 Hz, 2H), 7.60 (t, *J*=7.6 Hz, 2H), 7.49 (m, 1H), 7.42–7.28 (m, 7H), 6.45 (s, 1H), 3.24–3.15 (m, 4H), 2.59 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 158.9, 150.4, 141.0, 137.0, 132.4, 130.0, 128.7, 128.6, 128.5, 128.4, 127.5, 126.2, 124.4, 124.2, 119.6, 102.4, 34.2, 30.4, 21.5. Anal. Calcd for $C_{23}H_{20}O$: C, 88.43; H, 6.45. Found: C, 88.50; H, 6.46.

*4.3.52. 5-Fluoro-7-((4-methoxyphenyl)ethynyl)-2-phenylbenzo[b]furan (**7b**).* Yellow solid. Mp: 131–133 °C. MS: *m*/*z* 365 (MNa^+); IR (KBr): 2937, 2212, 1597, 1568, 1510, 1462, 1250, 1026 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.92 (d, *J*=7.6 Hz, 2H), 7.64 (d, *J*=8.4 Hz, 2H), 7.51–7.39 (m, 3H), 7.23–7.18 (m, 2H), 7.00–6.96 (m, 3H), 3.88 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 160.1, 158.8 (d, J_{C-F}=237 Hz), 157.6, 151.0, 133.4, 129.8, 129.1, 128.8, 125.2, 114.9, 114.8, 114.5, 114.1, 108.5, 106.7 (d, J_{C-F}=25 Hz), 101.6 (d, J_{C-F}=4 Hz), 94.9, 81.8, 55.3. Anal. Calcd for $C_{23}H_{15}FO_2$: C, 80.69; H, 4.42. Found: C, 80.78; H, 4.43.

*4.3.53. 7-((4-Methoxyphenyl)ethynyl)-5-methyl-2-phenylbenzo[b]furan (**7c**).* Yellow solid. Mp: 102–104 °C. MS: *m*/*z* 361 (MNa^+); IR (KBr): 2920, 2359, 1595, 1568, 1508, 1444, 1246, 1020 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.95 (d, *J*=7.2 Hz, 2H), 7.66 (d, *J*=8.8 Hz, 2H), 7.50 (t, *J*=7.6 Hz, 2H), 7.42–7.38 (m, 1H), 7.34 (d, *J*=6.4 Hz, 2H), 6.99–6.97 (m, 3H), 3.88 (s, 3H), 2.48 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 159.8, 156.4, 153.1, 133.3, 132.5, 130.4, 129.5, 128.8, 128.7, 125.1, 121.2, 115.6, 114.1, 107.2, 101.3, 93.7, 83.0, 55.3, 21.2. Anal. Calcd for $C_{24}H_{18}O$: C, 85.18; H, 5.36. Found: C, 85.25; H, 5.37.

*4.3.54. 2-(4-Methoxyphenyl)-5-methyl-7-(phenylethynyl)benzo[b]furan (**7d**).* White solid. Mp: 142–144 °C. MS: *m*/*z* 361 (MNa^+); IR (KBr): 2918, 2214, 1614, 1589, 1508, 1444, 1254, 1039 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.87 (d, *J*=8.4 Hz, 2H), 7.71 (d, *J*=6.4 Hz, 2H), 7.44–7.41 (m, 3H), 7.34 (s, 1H), 7.30 (s, 1H), 7.01 (d, *J*=8.8 Hz, 2H), 6.84 (s, 1H), 3.88 (s, 3H), 2.47 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 160.1, 156.6, 152.9, 132.4, 131.8, 129.8, 128.4, 128.3, 125.5, 126.6, 123.5, 123.2, 121.2, 114.2, 106.6, 99.6, 93.4, 84.4, 55.4, 21.2. Anal. Calcd for $C_{24}H_{18}O_2$: C, 85.18; H, 5.36. Found: C, 85.27; H, 5.37.

*4.3.55. 4-(2-(4-Methoxyphenyl)-5-methylbenzo[b]furan-7-yl)-2-methylbut-3-yn-2-ol (**7e**).* Yellow solid. Mp: 102–103 °C. MS: *m*/*z* 343 (MNa^+); IR (KBr): 3448, 2924, 2360, 1614, 1587, 1508, 1464, 1250, 1024 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.82 (d, *J*=8.8 Hz, 2H), 7.29 (s, 1H), 7.16 (s, 1H), 7.00 (d, *J*=8.4 Hz, 2H), 6.82 (s, 1H), 3.88 (s, 3H), 2.42 (s, 3H), 2.29 (s, 1H), 1.74 (s, 6H); ^{13}C NMR ($CDCl_3$): δ 160.1, 156.5, 153.0, 132.2, 129.7, 128.4, 126.7, 123.1, 121.1, 115.6, 114.2, 105.9, 99.5, 65.8, 55.4, 31.6, 21.1. Anal. Calcd for $C_{26}H_{22}O_3$: C, 78.73; H, 6.29. Found: C, 78.65; H, 6.28.

*4.3.56. 4-(5-Fluoro-2-phenylbenzo[b]furan-7-yl)-2-methylbut-3-yn-2-ol (**7f**).* Yellow solid. Mp: 93–95 °C. MS: *m*/*z* 317 (MNa^+); IR (KBr): 3354, 2927, 1599, 1473, 1444 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.88 (d, *J*=7.6 Hz, 2H), 7.49 (t, *J*=7.6 Hz, 2H), 7.42 (d, *J*=6.8 Hz, 1H), 7.21 (d, *J*=8.4 Hz, 1H), 7.07 (d, *J*=9.6 Hz, 1H), 6.98 (s, 1H), 2.35 (br s, 1H), 1.75 (s, 6H); ^{13}C NMR ($CDCl_3$): δ 158.6 (d, J_{C-F}=237 Hz), 157.9, 151.2, 140.3, 130.0, 129.1, 128.9, 125.1, 114.9 (d, J_{C-F}=27 Hz), 107.4 (d, J_{C-F}=27 Hz), 107.3, 101.6 (d, J_{C-F}=4 Hz), 99.3, 75.9, 65.8, 31.4. Anal. Calcd for $C_{24}H_{17}FO_2$: C, 77.54; H, 5.14. Found: C, 77.62; H, 5.15.

*4.3.57. 2-Methyl-4-(5-methyl-2-phenylbenzo[b]furan-7-yl)but-3-yn-2-ol (**7g**).* Yellow solid. Mp: 102–103 °C. MS: *m*/*z* 313 (MNa^+); IR (KBr): 3440, 2924, 2364, 1597, 1564, 1491, 1444 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.89 (d, *J*=7.6 Hz, 2H), 7.49–7.32 (m, 4H), 7.19 (s, 1H), 6.94 (s, 1H), 2.42 (s, 3H), 1.76 (s, 6H), 1.31 (br s, 1H); ^{13}C NMR ($CDCl_3$): δ 156.4, 153.2, 132.4, 131.1, 130.3, 129.4, 128.9, 128.8, 128.7, 125.0, 121.4, 106.2, 101.2, 98.2, 65.9, 31.6, 21.1. Anal. Calcd for $C_{25}H_{18}O_2$: C, 82.73; H, 6.25. Found: C, 85.65; H, 6.24.

*4.3.58. 1-Phenyl-1-(5-phenyl-7-(phenylethynyl)benzo[b]furan-2-yl) ethanol (**7h**).* Brown wax. MS: *m*/*z* 437 (MNa^+); IR (KBr): 3417, 2924, 2216, 1599, 1574, 1491, 1444 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.75–7.74 (m, 2H), 7.68–7.37 (m, 15H), 6.69 (s, 1H), 3.01 (br s, 1H), 2.09 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 162.9, 154.1, 145.0, 140.9, 136.8, 131.8, 129.0, 128.9, 128.6, 128.4, 128.3, 127.8, 127.6, 127.4, 127.2, 125.5, 123.1, 120.2, 107.7, 103.6, 94.0, 84.0, 73.5, 29.2. Anal. Calcd for $C_{30}H_{22}O_2$: C, 86.93; H, 5.35. Found: C, 86.85; H, 5.36.

*4.3.59. 1-(7-(Hept-1-ynyl)-5-phenylbenzo[b]furan-2-yl)-1-phenylethanol (**7i**).* Brown wax. MS: *m*/*z* 431 (MNa^+); IR (KBr): 3433, 2929, 2235, 1599, 1493, 1444 cm^{-1}; ^1H NMR ($CDCl_3$): δ 7.68–7.35 (m, 12H), 6.66 (s, 1H), 3.03 (s, 1H), 2.52 (t, *J*=6.8 Hz, 2H), 2.06 (s, 3H), 1.71–1.66 (m, 2H), 1.54–1.49 (m, 2H), 1.45–1.39 (m, 2H), 0.98

(t, J=6.8 Hz, 3H); ^{13}C NMR ($CDCl_3$): δ 162.5, 154.4, 145.1, 141.1, 136.7, 128.8, 128.7, 128.3, 128.1, 127.6, 127.4, 127.1, 125.5, 119.3, 108.5, 103.5, 95.6, 75.1, 73.5, 31.1, 29.2, 28.4, 22.3, 19.8, 14.1. Anal. Calcd for $C_{29}H_{28}O_2$: C, 85.26; H, 6.91. Found: C, 85.19; H, 6.92.

*4.3.60. (E)-2-(4-Methoxyphenyl)-5-methyl-7-styrylbenzo[b]furan (**9b**).* Yellow solid. Mp: 127–129 °C. MS: m/z 363 (MNa^+); IR (KBr): 2918, 1610, 1570, 1506, 1462, 1250, 1026 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.87 (d, J=8.8 Hz, 2H), 7.69 (d, J=8.4 Hz, 2H), 7.49–7.45 (m, 3H), 7.41–7.34 (m, 2H), 7.28–7.24 (m, 2H), 7.05–7.02 (m, 2H), 6.86–6.84 (m, 1H), 3.90 (s, 3H), 2.50 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 160.0, 156.1, 150.7, 137.9, 132.5, 131.4, 130.2, 128.8, 127.7, 126.7, 126.4, 124.1, 123.8, 123.5, 121.3, 119.9, 114.3, 99.6, 55.4, 21.4. Anal. Calcd for $C_{24}H_{20}O_2$: C, 84.68; H, 5.92. Found: C, 84.60; H, 5.93.

*4.3.61. (E)-5-Fluoro-2-phenyl-7-styrylbenzo[b]furan (**9c**).* Yellow solid. Mp: 103–105 °C. MS: m/z 337 (MNa^+); IR (KBr): 1595, 1495, 1471, 1446 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.91 (d, J=7.6 Hz, 2H), 7.67–7.63 (m, 3H), 7.54–7.37 (m, 7H), 7.17–7.14 (m, 2H), 7.00 (s, 1H); ^{13}C NMR ($CDCl_3$): δ 159.5 (d, J_{C-F}=237 Hz), 157.4, 148.7, 137.2, 132.8, 130.4, 130.1, 128.9, 128.2, 126.9, 125.0, 122.9, 122.8, 122.3, 110.0, 109.8 (d, J_{C-F}=26 Hz), 105.3 (d, J_{CF}=25 Hz), 101.5 (d, J_{C-F}=4 Hz). Anal. Calcd for $C_{22}H_{15}FO$: C, 84.06; H, 4.81. Found: C, 84.15; H, 4.82.

*4.3.62. 1-(2,5-Diphenylbenzo[b]furan-7-yl)piperidine (**10a**).* Yellow solid. Mp: 173–175 °C. MS (relative intensity): m/z 354 (M^+, 100); IR (KBr): 2944, 2931, 2811, 1637 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.93 (d, J=7.2 Hz, 2H), 7.70 (d, J=7.2 Hz, 2H), 7.51–7.48 (m, 4H), 7.42–7.38 (m, 3H), 7.07–7.05 (m, 2H), 3.48 (t, J=5.6 Hz, 4H), 1.95 (m, 4H), 1.75 (m, 2H); ^{13}C NMR ($CDCl_3$): δ 155.5, 146.3, 142.4, 138.3, 137.6, 130.8, 130.6, 128.8, 128.7, 128.5, 127.5, 126.8, 124.9, 112.3, 111.7, 102.0, 51.4, 27.0, 26.2, 24.6. Anal. Calcd for $C_{25}H_{23}NO$: C, 84.95; H, 6.56; N, 3.96. Found: C, 84.86; H, 6.57; N, 3.97.

*4.3.63. 4-(5-Methyl-2-phenylbenzo[b]furan-7-yl)morpholine (**10c**).* Yellow solid. Mp: 100–102 °C. MS: m/z 316 (MNa^+); IR (KBr): 3057, 2945, 2854, 1603, 1496, 1477, 1448 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.86 (d, J=7.6 Hz, 2H), 7.49 (t, J=7.6 Hz, 2H), 7.39 (t, J=7.2 Hz, 1H), 7.04 (s, 1H), 6.96 (s, 1H), 6.62 (s, 1H), 4.05 (t, J=4.4 Hz, 4H), 3.45 (t, J=4.4 Hz, 4H), 2.48 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 155.3, 144.9, 136.6, 133.3, 130.6, 128.8, 128.4, 124.8, 114.0, 112.6, 103.3, 101.6, 67.1, 50.4, 21.8. Anal. Calcd for $C_{19}H_{19}NO_2$: C, 77.79; H, 6.53; N 4.77. Found: C, 77.86; H, 6.54; N, 4.78.

*4.3.64. 1-(4-Fluorophenyl)-4-(5-methyl-2-phenylbenzo[b]furan-7-yl)piperazine (**10d**).* Yellow solid. Mp: 139–140 °C. MS: m/z 409 (MNa^+); IR (KBr): 2825, 1601, 1512, 1475, 1448 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.87 (d, J=7.6 Hz, 2H), 7.48 (t, J=7.6 Hz, 2H), 7.38 (t, J=7.6 Hz, 1H), 7.08–6.98 (m, 6H), 6.66 (s, 1H), 3.60 (t, J=4.8 Hz, 4H), 3.43 (t, J=4.8 Hz, 4H), 2.47 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 155.2, 145.0, 136.4, 133.3, 130.6, 130.5, 128.8, 128.4, 124.8, 118.2, 118.1, 115.6 (d, J_{C-F}=22 Hz), 114.0, 113.0, 101.6, 50.6, 50.0, 21.7. Anal. Calcd for $C_{25}H_{23}FN_2O$: C, 77.70; H, 6.00; N, 7.25. Found: C, 77.78; H, 5.99; N, 7.26.

*4.3.65. 1-Ethyl-4-(5-methyl-2-phenylbenzo[b]furan-7-yl)piperazine (**10e**).* Yellow solid. Mp: 83–84 °C. MS: m/z 321 (MH^+); IR (KBr): 2918, 2817, 1604, 1496, 1477, 1446 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.85 (d, J=7.6 Hz, 2H), 7.46–7.42 (m, 2H), 7.36 (t, J=7.6 Hz, 1H), 7.00 (s, 1H), 6.94 (s, 1H), 6.61 (s, 1H), 3.50 (t, J=4.4 Hz, 4H), 2.81 (t, J=4.4 Hz, 4H), 2.60 (q, J=7.2 Hz, 2H), 2.45 (s, 3H), 1.22 (t, J=7.2 Hz, 3H); ^{13}C NMR ($CDCl_3$): δ 155.1, 145.0, 136.6, 133.3, 130.7, 130.4, 128.7, 128.3, 124.7, 113.7, 112.8, 101.5, 52.9, 52.5, 49.8, 21.7, 11.9. Anal. Calcd for $C_{21}H_{24}N_2O$: C, 78.71; H, 7.55; N, 8.74. Found: C, 78.63; H, 7.56; N, 8.75.

*4.3.66. 1,1-Dimethyl-2-(5-methyl-2-phenylbenzo[b]furan-7-yl)hydrazine (**10f**).* Yellow solid. Mp: 112–114 °C. MS: m/z 267 (MH^+); IR (KBr): 2949, 2771, 1603, 1498, 1471, 1444 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.89 (d, J=7.6 Hz, 2H), 7.48 (t, J=7.6 Hz, 2H), 7.38 (t, J=7.6 Hz, 1H), 6.99 (s, 1H), 6.96 (s, 1H), 6.88 (s, 1H), 4.86 (s, 1H), 2.71 (s, 6H), 2.48 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 155.4, 141.6, 133.8, 132.2, 130.8, 129.4, 128.7, 128.3, 124.8, 111.1, 109.8, 101.8, 48.1, 21.8. Anal. Calcd for $C_{17}H_{18}N_2O$: C, 76.66; H, 6.81; N, 10.52. Found: C, 76.58; H, 6.82; N, 10.53.

*4.3.67. N-(5-Methyl-2-phenylbenzo[b]furan-7-yl)piperidin-1-amine (**10g**).* Yellow solid. Mp: 108–109 °C. MS: m/z 307 (MH^+); IR (KBr): 2941, 1604, 1495, 1469 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.92 (d, J=7.6 Hz, 2H), 7.52–7.40 (m, 3H), 7.04 (s, 1H), 6.97 (s, 1H), 6.88 (s, 1H), 5.02 (s, 1H), 2.86 (s, 4H), 2.51 (s, 3H), 1.86–1.80 (m, 4H), 1.53 (s, 2H); ^{13}C NMR ($CDCl_3$): δ 155.3, 141.6, 133.7, 132.3, 130.9, 129.4, 128.7, 128.2, 124.8, 110.9, 109.6, 101.9, 57.7, 26.2, 23.8, 21.9.

*4.3.68. 1-(5-Methyl-2-phenylbenzo[b]furan-7-yl)piperazine (**10h**).* Yellow solid. Mp: 120–122 °C. MS: m/z 293 (MH^+); IR (KBr): 3315, 3102, 2946, 2834, 1598, 767 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.85 (d, J=8 Hz, 2H), 7.46 (t, J=7.6 Hz, 2H), 7.36 (d, J=7.6 Hz, 1H), 7.0 (s, 1H), 6.94 (s, 1H), 6.61 (s. 1H), 3.39 (t, J=4.8 Hz, 4H), 3.19 (t, J=4.8 Hz, 4H), 2.45 (s, 3H), 1.95 (br s, NH); ^{13}C NMR ($CDCl_3$): δ 155.1, 145.1, 137.2, 133.2, 130.7, 130.5, 128.7, 128.3, 124.8, 113.7, 112.8, 101.5, 51.3, 46.3, 21.7. Anal. Calcd for $C_{19}H_{20}N_2O$: C, 78.05; H, 6.89; N, 9.58. Found: C, 78.13; H, 6.90; N, 9.59.

*4.3.69. 1-(5-Fluoro-2-phenylbenzo[b]furan-7-yl)piperidine (**10i**).* Yellow solid. Mp: 200 °C (dec). MS (relative intensity): m/z 295 (M^+, 100), 212 (10), 77 (9); IR (KBr): 2933, 1614, 1495, 1475 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.86 (d, J=7.6 Hz, 2H), 7.48 (t, J=7.6 Hz, 2H), 7.39 (t, J=7.6 Hz, 1H), 6.97 (s, 1H), 6.81 (dd, J_1=8.0 Hz, J_2=2 Hz, 1H), 6.53 (dd, J_1=12.0 Hz, J_2=2.0 Hz, 1H), 3.40 (t, 4H), 1.88–1.85 (m, 4H), 1.72–1.68 (m, 2H); ^{13}C NMR ($CDCl_3$): δ 160.0 (d, J_{C-F}=236 Hz), 156.1, 142.7, 138.8 (d, J_{CF}=104 Hz), 130.2, 130.0, 128.8, 128.6, 124.8, 101.9 (d, J_{C-F}=5 Hz), 100.0 (d, J_{C-F}=29 Hz), 97.7 (d, J_{C-F}=25 Hz), 51.0, 26.0, 24.5. Anal. Calcd for $C_{19}H_{18}FNO$: C, 77.27; H, 6.14; N, 4.74. Found: C, 77.20; H, 6.15; N, 4.75.

*4.3.70. 1-(2-Pentyl-5-phenylbenzo[b]furan-7-yl)piperidine (**10j**).* Yellow oil. MS: m/z 369 (MNa^+); IR (KBr): 2931, 1603, 1577, 1466 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.66 (d, J=7.6 Hz, 2H), 7.50–7.46 (m, 2H), 7.38–7.28 (m, 2H), 6.98 (s, 1H), 6.43 (s, 1H), 3.40–3.37 (m, 3H), 2.87–2.81 (m, 2H), 1.92–1.70 (m, 9H), 1.48–1.41 (m, 4H), 0.98 (t, J=6.8 Hz, 3H); ^{13}C NMR ($CDCl_3$): δ 159.4, 146.1, 142.6, 138.0, 136.9, 130.5, 128.7, 127.5, 126.6, 111.7, 110.5, 102.3, 51.4, 31.4, 28.4, 27.5, 26.2, 24.6, 22.5, 14.0. Anal. Calcd for $C_{24}H_{29}NO$: C, 82.95; H, 8.41; N, 4.03. Found: C, 82.87; H, 8.42; N, 4.04.

*4.3.71. 4-(2-Pentyl-5-phenylbenzo[b]furan-7-yl)morpholine (**10k**).* Yellow solid. Mp: 55–57 °C. MS: m/z 372 (MNa^+); IR (KBr): 2924, 1597, 1574, 1475, 1448, 1273, 1026 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.66 (d, J=7.2 Hz, 2H), 7.49 (t, J=7.6 Hz, 2H), 7.40–7.36 (m, 2H), 6.95 (s, 1H), 6.45 (s, 1H), 4.03 (t, J=4.4 Hz, 4H), 3.45 (t, J=4.4 Hz, 4H), 2.85 (t, J=7.2 Hz, 2H), 1.84–1.79 (m, 2H), 1.46–1.42 (m, 4H), 0.99 (t, J=7.2 Hz, 3H); ^{13}C NMR ($CDCl_3$): δ 159.7, 145.8, 142.4, 137.1, 136.7, 130.8, 128.7, 127.5, 126.8, 112.3, 109.9, 102.4, 67.1, 50.3, 31.4, 28.4, 27.5, 22.5, 14.1. Anal. Calcd for $C_{23}H_{27}NO_2$: C, 79.05; H, 7.79; N, 4.01. Found: C, 79.13; H, 7.80; N, 4.02.

*4.3.72. N-Isopropyl-2-pentyl-5-phenylbenzo[b]furan-7-amine (**10l**).* Yellow oil. MS: m/z 343 (MNa^+); IR (KBr): 2952, 1604, 1577, 1468, 1448 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.74 (d, J=7.2 Hz, 2H), 7.50 (t,

J=7.6 Hz, 2H), 7.40–7.39 (m, 2H), 7.23 (s, 1H), 6.45 (s, 1H), 4.87 (s, 1H), 2.83 (t, J=7.2 Hz, 2H), 2.71 (s, 6H), 1.85–1.82 (m, 2H), 1.49–1.46 (m, 4H), 1.01 (t, J=6.8 Hz, 3H); ^{13}C NMR ($CDCl_3$): δ 159.7, 142.7, 142.3, 137.4, 132.5, 129.6, 128.6, 127.6, 126.6, 109.3, 107.2, 102.6, 48.0, 31.5, 28.5, 27.5, 22.5, 14.1. Anal. Calcd for $C_{22}H_{27}NO$: C, 82.20; H, 8.47; N, 4.36. Found: C, 82.28; H, 8.46; N, 4.37.

*4.3.73. N-(2-Pentyl-5-phenylbenzo[b]furan-7-yl)piperidin-1-amine (**10m**).* Yellow oil. MS: m/z 363 (MH^+); IR (KBr): 2933, 1604, 1577, 1469, 1442 cm^{-1}; ^{1}H NMR ($CDCl_3$) δ: 7.75 (d, J=7.6 Hz, 2H), 7.52 (t, J=7.6 Hz, 2H), 7.42–7.39 (m, 2H), 7.22 (s, 1H), 6.46 (s, 1H), 5.03 (s, 1H), 2.87–2.83 (m, 6H), 1.87–1.81 (m, 6H), 1.54–1.43 (m, 6H), 1.03 (t, J=6.8 Hz, 3H); ^{13}C NMR ($CDCl_3$) δ: 159.6, 142.9, 142.4, 137.3, 132.5, 129.6, 128.6, 127.7, 126.6, 109.2, 107.1, 102.6, 57.6, 31.5, 28.6, 27.6, 26.2, 23.8, 22.5, 14.1. Anal. Calcd for $C_{24}H_{30}N_2O$: C, 79.52; H, 8.34; N, 7.73. Found: C, 79.45; H, 8.35; N, 7.74.

*4.3.74. 1-(4-Fluorophenyl)-4-(2-pentyl-5-phenylbenzo[b]furan-7-yl) piperazine (**10n**).* Yellow wax. MS: m/z 465 (MNa^+); IR (KBr): 2929, 1603, 1508, 1473, 1450 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.68 (d, J=7.2 Hz, 2H), 7.50 (t, J=7.6 Hz, 2H), 7.41–7.38 (m, 2H), 7.09–7.02 (m, 5H), 6.47 (s, 1H), 3.61 (t, J=4.8 Hz, 4H), 3.42 (t, J=4.8 Hz, 4H), 2.87 (t, J=7.2 Hz, 2H), 1.86–1.83 (m, 2H), 1.48–1.34 (m, 4H), 1.00 (t, J=6.8 Hz, 3H); ^{13}C NMR ($CDCl_3$): δ 159.7, 158.4 (d, J_{C-F}=237 Hz), 148.1, 145.9, 142.4, 137.1, 136.6, 128.8, 127.6, 126.8, 118.2, 118.1, 115.6 (d, J_{C-F}=22 Hz), 112.4, 110.3, 102.4, 50.5, 50.0, 31.4, 28.5, 27.5, 22.5, 14.1. Anal. Calcd for $C_{29}H_{31}FN_2O$: C, 78.70; H, 7.06; N, 6.33. Found: C, 78.62; H, 7.05; N, 6.34.

*4.3.75. 4-(2-(4-Methoxyphenyl)-5-methylbenzo[b]furan-7-yl)morpholine (**10o**).* Yellow solid. Mp: 129–130 °C. MS: m/z 346 (MNa^+); IR (KBr): 2850, 1601, 1568, 1508, 1479, 1444, 1250, 1026 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.77 (d, J=8.4 Hz, 2H), 7.00–6.98 (m, 3H), 6.81 (s, 1H), 6.58 (s, 1H), 4.03 (s, 4H), 3.88 (s, 3H), 3.42 (s, 4H), 2.45 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 159.9, 155.4, 144.6, 136.4, 133.2, 130.8, 126.2, 123.4, 114.2, 113.9, 112.1, 99.9, 67.1, 55.4, 50.4, 21.8. Anal. Calcd for $C_{20}H_{21}NO_3$: C, 74.28; H, 6.55; N, 4.33. Found: C, 74.36; H, 6.56; N, 4.34.

*4.3.76. 1-(7-(2,2-Dimethylhydrazinyl)-5-phenylbenzo[b]furan-2-yl)-1-phenylethanol (**10p**).* Yellow wax. MS: m/z 373 (MH^+); IR (KBr): 3427, 2924, 1603, 1574, 1495, 1446 cm^{-1}; ^{1}H NMR ($CDCl_3$): δ 7.68 (d, J=7.6 Hz, 2H), 7.53–7.21 (m, 10H), 6.68 (s, 1H), 4.84 (s, 1H), 2.85 (br s, 1H), 2.63 (s, 6H), 2.01 (s, 3H); ^{13}C NMR ($CDCl_3$): δ 161.2, 145.3, 142.6, 142.4, 137.9, 137.7, 132.8, 132.6, 128.6, 128.3, 127.6, 126.7, 125.4, 109.7, 108.0, 104.2, 73.5, 47.9, 29.7, 29.4. Anal. Calcd for $C_{24}H_{24}N_2O_2$: C, 77.39; H, 6.49; N, 7.52. Found: C, 77.48; H, 6.50; N, 7.53.

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